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**The Rise of the Customer Success Function as Catalyst for Team Innovation:
The Importance of Boundary Spanning, Knowledge Integration, and Shared Vision**

A Thesis

Submitted to the Faculty

of

Drexel University

by

André Wagner

in partial fulfillment of the

requirements for the degree

of

Doctorate in Business Administration

February 2021



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ABSTRACT

The Rise of the Customer Success Function as Catalyst for Team Innovation: The Importance of Boundary Spanning, Knowledge Integration, and Shared Vision

André Wagner

Drexel University, February 2021

Chairpersons: Daan van Knippenberg, Lauren D’Innocenzo

In this study, I assess the causal impact of boundary spanning on team innovation. I explore conditions and processes in 144 engagement teams in an enterprise software company driving innovation with customers in North America across industries. In this context, innovation is defined as the development and implementation of technology-based ideas in organizations. Using an integrated conditional process analysis, I find that boundary-spanning team practices combined with a shared vision and functional background diversity lead to a significant team innovation increase. I further assess the team process of information elaboration through which functional diversity and boundary spanning foster solution use. In particular, I argue that boundary spanning amplifies the effects of shared vision and information elaboration, leading to significantly increased team innovation. Consistent with this argument, I find that functional diversity's positive impact on team innovation through information elaboration is conditional on a shared vision, whereas the impact is more significant when team members share a commitment to common goals for their work. Therefore, my findings contribute actionable insight into how to enable cross-functional engagement teams to successfully drive customer innovation with enterprise software in the technology industry.

CHAPTER 1: INTRODUCTION

Context

New digital technologies and business models are radically transforming our economies and reshaping industry sectors around the world. Digital offerings from dominant technology companies including Amazon, Apple, Facebook, Google, and Microsoft are indivisible from most people's lives in Western societies. Such digital platform businesses – even relatively smaller companies like Airbnb with about 6,000 employees and a market cap of \$74 billion – challenge incumbent enterprises, such as Hilton with 173,000 employees and \$30B market capitalization (Yahoo!finance, 2020), in unprecedented ways and achieve higher market valuations with significantly fewer resources than traditional businesses (Cusumano, Gawer, & Yoffie, 2019). Therefore, for many businesses, the question is no longer whether they need to innovate but how.

One of the strategic imperatives for this digital business transformation is cloud computing (Brynjolfsson, Hofmann, & Jordan, 2010). In the second decade of the cloud, the role of software has moved beyond supporting the business. Digital transformation is the business (Siebel, 2017). Disney, for example, created Disney Plus, a completely new direct-to-consumer digital business model. While the total 2020 Disney revenue was down 6 percent because of the COVID-19 pandemic with theme park closures and the shift of many movies to 2021, Disney Plus counted 86+ million global subscribers in its first year, driving 41 percent revenue growth in its segment. Disney is now restructuring its media and entertainment divisions to accelerate its digital strategy by centralizing its delivery units with Disney Plus. However, Disney also acknowledges that it will not be profitable until 2024 because of additional investment requirements (Zeitchik, 2020).

In this context, it is no surprise that lack of resources and knowledge have been identified as a critical challenge for businesses innovating with cloud infrastructure, platform, and application services (Retana, Forman, Narasimhan, & Niculescu, 2018). Specifically, Lucas, Swanson, and Zmud (2007) asked for studies that help us understand better how innovation with digital technologies can lead to capabilities that provide sustainable competitive advantages.

Problem Statement

With the rise of subscription business models for technology companies, it is no longer good enough to sell enterprise application software packages; instead, technology providers need to ensure that customers adopt their solutions and that the expected value is delivered. In the traditional software license model, the software and what was accomplished with it was the customer's responsibility. After an up-front payment, most revenue and profit were guaranteed for the provider regardless of whether the customer received value or not. Today, technology providers realize that sustainable business success requires an organizational culture, determination, and strategic vision that prioritizes enduring customer value (Kumar & Reinartz, 2016). In other words, the provider needs to ensure value realization for the customer and orchestrate a simplified customer journey that ensures ongoing satisfaction and an optimal experience as recurring revenue streams could be stopped at any point if the service does not meet the needs. In response to this challenge, technology providers are developing customer success functions to drive adoption and usage (Mehta & Pickens, 2020). Lah, Nanus, and Stegman (2017) define customer success as a technology provider's function responsible for ensuring that customers receive the full value and are on track to achieve their desired

business outcomes with the acquired technologies. As shown in Figure 1, the customer success function complements the sales function to ensure full life cycle engagement and protect renewal revenues. Renewals are critical for long-term profitability. In fact, studies by Reichheld and Schefter (2000) found that a “5% improvement in retention increases profits by up to 95%.”

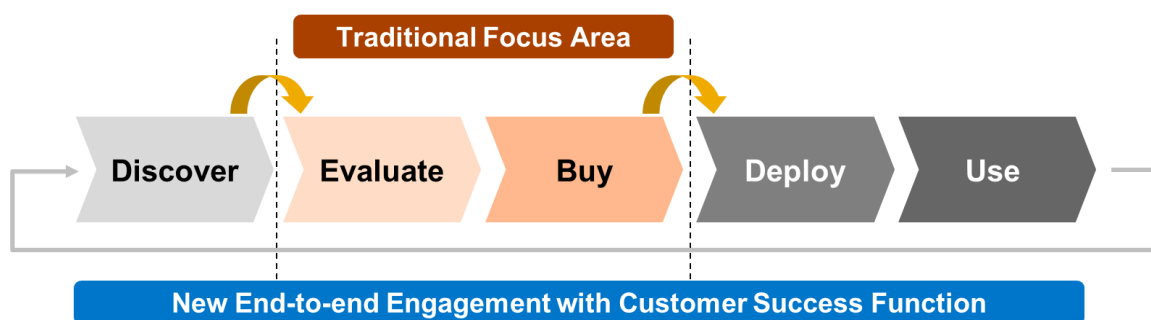


Figure 1 Customer Journey Support by Technology Providers

Nevertheless, as an emerging function, there has not been a reliable and proven model for technology providers to efficiently drive customer success at scale (Mehta & Pickens, 2020). Despite resource investments, technology providers still struggle with subscription renewals and customer business innovation. The complementary deployment of customer success resources, for example, could even lead to decreasing customer lifetime value. Despite the importance of churn management and the extensive academic work on customer churn and retention in the field of marketing, information systems, and operations (e.g., Hadden, Tiwari, Roy, & Ruta, 2007; Neslin, Gupta, Kamakura, Lu, & Mason, 2006), there is a gap whether it is efficient for businesses to target customers at risk without knowing if or to what intervention efforts they are responsive (Ascarza, 2018). Ascarza, Iyengar, and Schleicher (2016), for example, found that proactive

retention campaigns can increase rather than decrease customer churn. Digital transformation studies from Forth, Reichert, de Laubier, and Chakraborty (2020) and McKinsey (2018), furthermore, found that less than one-third of organizational transformations succeed at improving performance and sustaining it over time.

The research of Anderson, Potocnik, and Zhou (2014), van Knippenberg (2017), and West (2002) highlights that understanding factors that enhance implementation success deserves more focus since it is more challenging than idea generation. Bringing creative ideas to life in organizations requires executive buy-in and the mobilization of resources and support, change management, diligence, persistence, and commitment (Drucker, 2002). Anderson and colleagues (2014) summarize in their state-of-the-science review that innovation is more critical than ever before for organizational survival and corporate growth. Moreover, it is even more relevant in today's knowledge-intensive and fast-paced business environment with digital competitors and changing customer needs and regulations (van Knippenberg, 2017). However, the conditions that foster team innovation are not very well understood (Hoever, van Knippenberg, van Ginkel, & Barkema, 2012). While the creativity aspect of innovation has received much attention in research, the contingencies of the relationship between ideation and implementation have been identified as a critical understudied issue (van Knippenberg, 2017).

Small and large businesses across industries facing the need to innovate rely on teams as a predominant way of organizing work (Mathieu, Hollenbeck, van Knippenberg, & Ilgen, 2017). Although teams play a central role in the contemporary business environment (Barczak, Griffin, & Kahn, 2009; Resick, Murase, Randall, & DeChurch, 2014), innovation research has mainly focused on individual or organizational level of

analysis and has provided inconclusive results regarding crucial factors that give a team informational resources, and what motivates a team to use those resources to drive innovation (Hülshager, Anderson, & Salgado, 2009; Lucas et al., 2003; Shams, Vrontis, Weber, & Tsoukatos, 2018; Tzabbar & Vestal, 2018; van Knippenberg, 2017).

Furthermore, the extant learning literature has shown the role of organizational learning in overcoming knowledge barriers but much less is known about how technology providers' customer success strategies affect the consumption of cloud services (Retana et al., 2018). The pressing question arising from these insights puts the spotlight on the mechanisms and conditions that facilitate knowledge integration. This leads to my research questions.

Research Questions

Firstly, I address the question, *how and under what conditions are customer engagement teams in technology companies likely to benefit from diverse informational resources in pursuing innovation challenges?* The central thesis is that team innovation benefits from informational resources through its relationship with information elaboration and engenders team innovation success only if teams operate in a positive climate for innovation. Factors contributing to a team's informational resources, such as functional background diversity and social networks, are beneficial for team innovation. Furthermore, Hoever and colleagues (2012) and Drach-Zahavy and Somech (2001) show that knowledge exchange and integration are conducive to team innovation. Teams should be more innovative the more team members bring different perspectives and can create novel insight by engaging in information elaboration to address innovation challenges. However, team innovation benefits from informational resources only if the

team engages in information elaboration. Even if a team possesses informational resources, it is not automatically given that the team uses them. Edmondson and Harvey (2018) also found positive and negative effects of diversity. The equivocal results of team diversity point to the presence of moderators, which I study by determining when functional diversity is beneficial to team innovation. In that regard, van Ginkel and van Knippenberg (2008) suggested that the team climate for innovation positively moderates the relationship between team informational resources and information elaboration. This leads me to my next question.

Secondly, I address the question, *what is the role of boundary spanning, and how can it stimulate team innovation success for customer engagement teams?* Building on the work of many innovation researchers who highlight the need for organizations to manage innovation within and outside the firm to sustain their competitive advantage (Alexander & van Knippenberg, 2014; Gawer & Cusumano, 2014; Leifer, McDermott, O'Connor, Peters, Rice, & Veryzer, 2000), the notion of boundary spanning is straight forward: Teams with members who promote cross-boundary work, in the sense of interface collaboration spanning functions or domains, significantly outperform teams without such people (Casciaro, Edmondson, & Jang, 2019). This effect is specifically relevant for teams working complex service engagements and pursuing innovations that require tight business and information technology (IT) alignment (Henderson & Venkatraman, 1993; Sabherwal, Hirschheim, & Goles, 2001). Kleinbaum and Tushman (2007) further argue that collaboration across organizational units is necessary for innovations to be successful.

Tambe, Hitt, and Brynjolfsson (2012) found that external focus, decentralization, and IT investments are correlated and demonstrated that firms that combine all three practices derive substantially greater benefits from their IT investments. However, on the team level, the empirical evidence of boundary spanning success is equivocal, some research indicating a facilitating effect and others finding an inhibiting effect to knowledge exchange (Tortoriello, Reagans, & McEvily, 2012). Ancona and Caldwell (1992), for example, found that cross-functional teams that engage in external communication drive long-term team innovation but also pointed out that specific types of boundary spanning such as too much scouting or pure ambassadorial activities without task-coordination are related to lower performance. Boundary-spanning practices in the IT industry, generally speaking, help to break down silos (Casciaro et al., 2019) and reduce the risk of being “too creative,” in the sense of solely focusing on idea generation without or little thought about planning, prioritization, feasibility, resource mobilization, solution adoption, and change management. The integration of information across silos helps teams drive business and technology alignment necessary for sustainable innovation success (Reynolds & Yetton, 2015).

Purpose

My research aims to understand better the conditions that foster team innovation. Specifically, developing a deeper level of understanding under what circumstances customer engagement teams successfully drive customer innovation with software solutions. Team innovation research will benefit from integrating the knowledge and team climate perspectives (van Knippenberg, 2017). In this study, I conceptually apply van Knippenberg’s (2017) model of team innovation shown in Figure 2 that flows from

integrating these perspectives in the combined field of business / IT alignment and customer retention.

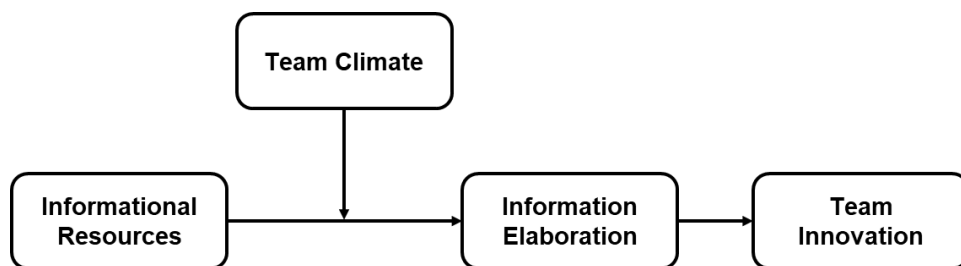


Figure 2 Knowledge Integration / Team Climate Model of Team Innovation

By building on van Knippenberg's state-of-the-science integrated process model and including the notion of boundary spanning, I seek to further shed light on idea implementation and study team innovation not only from the perspective of knowledge integration and team climate but also from the perspective of the external context in which engagement teams operate to bring creative ideas to life and drive business outcomes for the customer. Specifically, I study the impact of functional background diversity on team innovation through the team process of information elaboration motivated by a shared vision and boundary spanning.

The causal impact of boundary-spanning practices on team innovation and the contingencies inherent in the knowledge integration and team climate model is empirically assessed in an applied research approach with a cross-sectional company survey that is combined with field experiment data in the context of complex technology services engagements of a global enterprise software company, hereinafter referred to as the company, and its customers. In doing so, I seek to advance information elaboration theory by integrating perspectives and developing practical guidance for innovation

management, addressing both theoretical and practical calls to action from Anderson and colleagues (2014) and van Knippenberg (2017) and further adding rigor to business issues around the emerging customer success functions of technology providers and developing a deeper level of understanding under what processes and conditions customer engagement teams drive customer innovation with enterprise software technology solutions.

In summary, the analysis of mechanisms that facilitate team innovation will provide insights for three groups of stakeholders. First, the study contributes to the organizational behavior research community by addressing the research gap regarding the integration of knowledge and team climate perspectives. Second, since the deployment of cross-functional customer engagement teams is a strategic decision with implications on cost structures and profitability, the findings are relevant for technology firms that engage collaboratively with their customers to develop effective go-to-market strategies. Third, these insights are of interest to organizations working with co-innovation partners on their digital strategies and seek to optimize their engagement model. Therefore, research of the circumstances conducive to team innovation is imperative for business and IT practitioners as well as organizational behavior, leadership, and organizational psychology academia. In the following section, I summarize an example engagement to illustrate the work of customer engagement teams in an actual, real-life situation.

An Example from the Construction Industry

After the mining downturn in 2012, an equipment company faced a changing market environment resulting in business contraction that required cost-cutting measures and improvements in efficiency across the business. The human resources (HR) organization

decided they needed to improve the way they worked in order to create a sustainable business. As a trusted partner in the HR transformation, the technology provider company's customer engagement team helped this organization successfully migrate largely paper-based HR processes into an integrated HR platform giving the business increased transparency and enabling insight-driven decisions. Such an HR transformation requires executive support, business process adjustments, user training, organizational change management, measurement of improvements, and ongoing governance from the business perspective and data quality management, system integration, security management, and support on the technical side. All such activities were planned, coordinated, and executed by the engagement team in collaboration with this company. As a result, the company has significantly improved HR processes' efficiency, reduced manual, time-consuming administrative work, increased employee retention, reduced lead times to fill open positions through automated HR processes, and repositioned HR from an administrative to a strategic business function. The chief human resource officer explains that *“we see a shift now in how the HR team is seen by the business and by the board. We are no longer being seen as a cost to the business or back-office support. There is being a shift because of the efficiencies we gain now with the new operating model. Now we are much more seen as a partner for the generation of revenue and the generation of profit as opposed to just the administrative cost center.”*

Assumptions

There are several assumptions and restrictions that influence my research objectives, the literature review, and the methodological research design. First and foremost, the findings will only represent teams from one technology provider company

operating with customers in the U.S. and Canada. Unlike prior team research studies, which often analyzed teams in educational or health care settings with perceptual outcome measures (Drach-Zahavy & Somech, 2001; Fay, Borrill, Amir, Haward, & West, 2006), I analyze cross-functional customer engagement teams working with customers across industries with an objective measurement method for team innovation. Regarding the experimental part of the research design, it must be noted that despite the company's best efforts and an initial random assignment between the treatment and control group, not all selected customers did agree to program participation. Therefore, the company experiment's design, which in part I build my data analysis on, is considered a quasi-experimental study design with minor imperfections that often come with real-life customer engagement situations. And then, of course, there is the global COVID-19 crisis, which impacted the second half of the company experiment. I included these aspects in the customer control conditions to ensure that it does not bias results. Furthermore, the company experiment focused on existing customers only and did not include engagement situations with prospects. For a detailed description, please see the section data and sample in the methods chapter.

Finally, while I am focusing on the two main themes of knowledge integration and team climate in management and organizational behavior research, the broad field of team creativity and innovation cannot be reduced entirely to these two perspectives only. Other perspectives and topics, such as team leadership, team composition, cross-cultural differences, and team dynamics, are outlined in the future research section.

CHAPTER 2: LITERATURE REVIEW

Theoretical Background

This study builds on the social categorization theory that takes similarities and differences as a basis for categorizing individuals into groups and focuses on relational aspects of group processes (Turner et al., 1987 cited in van Knippenberg, De Dreu, & Homan, 2004). Van Knippenberg and colleagues (2004), building on this theory, developed the categorization-elaboration model (CEM) that integrates social categorization perspectives with decision making in team elaboration and highlights the importance of diversity implications for team performance. Diversity is defined as *“differences between individuals on any attribute that may lead to the perception that another person is different from self”* (van Knippenberg et al., 2004: 1008). I am building on the belief in the value of diversity for team innovation (van Knippenberg, Nishii, & Dwertmann, 2020). However, an objective analysis of the available evidence in the diversity literature suggests that it does not consistently engender team creativity and innovation (e.g., Edmondson & Harvey, 2018). Van Knippenberg and Schippers (2007) support this conclusion and outline that diversity can not only enhance but also impede team performance, primarily through the adverse effects associated with social categorization processes for demographic diversity.

Building on diversity theory, I explore the effects of moderators by determining under which conditions diversity is beneficial for team innovation. The CEM has been proven as a useful framework to study such moderating influences in an integrated model of diversity and team innovation (van Knippenberg & Hoeve, 2018). In this study, I focus specifically on functional background diversity since cross-functional teams are

favorable for digital business innovation because of their capacity to engage in complex problem solving (Resick et al., 2014). Functional diversity refers to differences between group members in their functional background.

Customer engagement teams are defined as virtual work groups of people with different functional expertise sharing the responsibility for group-level tasks to explore new business opportunities and apply enterprise software and technology solutions by engaging with their customers (Kozlowski & Bell, 2003). Exploration includes problem identification focused on customer needs, ideation of solutions that address the needs and problems, and eventually the implementation of those solutions to address the identified customer needs (March, 1991). Within technology providers, such teams of knowledge workers are typically referred to as account teams and comprise people with cross-functional backgrounds. Now increasingly, account teams include team members performing customer success functions, such as deployment planning, feature adoption, user training, guidance for optimal solution usage, support, ongoing system health monitoring, and results tracking (Lah et al., 2017; Mehta & Pickens, 2020). These cross-functional teams include members with different functional backgrounds, such as marketing, HR, logistic, and IT.

The following literature review will discuss the most recent and relevant published research and is grounded on the literature review by van Knippenberg (2017) and van Knippenberg and Hoeve (in press) that are arguably the most eminent reviews of team creativity and innovation literature in the recent history. Van Knippenberg (2017) identified the knowledge integration perspective and the team climate perspective as the main themes for team innovation research. I begin my literature review with an overview

of team innovation and then focus on the team process of information elaboration in connection to the available research about functional background diversity by exploring the knowledge integration perspective and highlighting the importance of boundary spanning. Then I will review the shared vision literature and further explore the importance of the team climate perspective for information elaboration by focusing on high-quality published research in management and organizational behavior. While reviewing the literature in the following sections, I will also develop my hypotheses and conclude this chapter with a summary of my research model.

Team Innovation

Innovation is defined as the development and implementation of ideas in an organization (Edmondson, 2003). Within an organization, teamwork brings together different resources, expertise, and perspectives to solve complex business problems (Thayer, Petruzzelli, & McClurg, 2018). Following West and Farr (1990), I further define team innovation as a team's capability to generate technology-based ideas new for the adopting customer and put these ideas into practice such that they achieve the desired business outcomes. It is essential not to confuse innovation with invention. For team innovation, it is not required to generally create something new but rather the intentional introduction and application of enterprise software and technology solutions that are new for the adopting customer and are designed to improve processes, products, or procedures within that organization (West & Farr, 1990: 9). Such ideas, processes, or products could already exist outside the adopting organization (van Knippenberg, 2017). In fact, since the unit of analysis in this study is the customer engagement team, most innovations of such teams do exist and have been already implemented in similar forms in other

organizations. That is why many practitioners in the field seek engagement with technology providers to benefit from such insight that is often regarded as best practices. The need and importance of best practices are also evident in the role of communities, such as the Technology Services Industry Association, and analyst firms, such as Gartner or International Data Corporation, that provide IT advice and research for leaders across industries to achieve their goals and optimize their businesses. The development, documentation, and sharing of such best practices help drive progress in the entire technology industry and specifically speak to research calls for action addressing the importance of idea implementation (Anderson et al., 2014; van Knippenberg, 2017). Ultimately, it is only through the successful implementation of ideas and concepts that the desired business outcomes are achieved (West, 2002).

In team creativity and innovation research, idea implementation is often implied and combined with idea development. Apart from brainstorming, which is geared towards idea generation specifically, most research assumes that creative and innovative ideas somehow automatically lead to innovation, conceiving both creativity and innovation as integral parts of the same process. In that respect, Alexander and van Knippenberg (2014) highlight that teams need to mobilize resources and collaborate with stakeholders outside the team to drive innovation success. They specifically identify the roles of ambidextrous leadership and reflexive team processes as crucial success factors for radical innovation. All of which take dedicated team efforts and by no means happen by themselves. From that perspective, the study of Somech and Drach-Zahavy (2013) is a good example that differentiates phases of creativity and implementation by studying team innovation in primary care teams.

For IT-based innovations to become successfully adopted by an organization, it is essential to realize that critical challenges are often beyond the team and typically require executive support and collaboration across divisions, such as HR and IT (Leifer et al., 2000). To rise to the challenge of adopting IT innovations, Swanson and Ramiller (1997) proposed the conceptual framework of an “*organizing vision*” that captures what a specific innovation is suitable for, how it works, and how it should be implemented. Through such a framework and the related community discourse, teams can better translate IT-related buzzwords, such as artificial intelligence, into a real understanding of new IT, including its requirements, design, applicability, and use. Charting a successful IT implementation requires not only new organizational designs but also the continuous reflection of own experiences and the evolving application in the industry (Swanson & Ramiller, 1997). All of which creates an intensive need for knowledge integration. Therefore, cross-boundary teamwork became a prevalent practice to achieve innovation success (Edmondson & Harvey, 2018). Moser, Dawson, and West (2019), for example, found that information sharing and helping were positively related to innovation.

In information systems research, it is well-documented that besides a firms’ internal capabilities, executive support, empowerment, operational excellence, and culture, it is the technical know-how that affects the timing of the adoption of a new information system and the extent of usage and diffusion after the initial stages (Seddon, Calvert, & Yang, 2010). Therefore, beginning with the knowledge integration and management perspective works well with the vast organizational behavior literature and aligns well with research in management information systems (Alegre, Sengupta, & Lapiedra, 2011; Andreeva & Kianto, 2012).

Knowledge Integration Perspective

By applying the knowledge integration lens, team innovation is seen to follow from integrating diverse information, ideas, perspectives, and knowledge. Bell, Villado, Lukasik, and Briggs (2011) and van Knippenberg, van Ginkel, and Homan (2013) argue that having a greater variety of functional backgrounds in a team is beneficial for team innovation because such teams can utilize different pools of information, knowledge, and experiences. Different perspectives can then lead to discussion and better integration of information in a collaborative team process resulting in a deeper understanding of the problem, finding a broader range of potential solutions, and alternative implementation approaches, ultimately resulting in increased team innovation. This logic is supported by meta-analyses of van Dijk, van Engen, and van Knippenberg (2012).

Team Diversity

Diversity research has a long history, and already over three decades ago, Bantel and Jackson (1989) found that functional diversity in top management teams predicted organizational innovation. In 1992, Ancona and Caldwell (1992) presented their research of boundary spanning practices in research and development (R&D) teams demonstrating that functional diversity enhances team innovation through team external communication. Ancona and Caldwell (1992) defined boundary spanning as external communication, including ambassadorial, scouting, and task-coordinator activities. Ambassadorial activities include accessing and mobilizing critical resources and persuading others to support the team's goals. Scouting involves scanning for ideas and gathering information about technologies, competition, and the industry. Coordination work embraces activities to create alignment with the customer and establish a feedback loop.

Later, more supporting evidence for diversity's main effects was found confirming their research. Tzabbar and Vestal (2015), also studying R&D teams, found a curvilinear relationship between geographic dispersion and novelty of innovation that was moderated by relational strength and status asymmetry. However, research on boundary spanning was long evolving in two parallel, almost independent research streams (Schotter, Mudambi, Doz, & Gaur, 2017). One stream studies boundary spanning as an organizational function (Zhao & Anand, 2013), and the other focusing on individuals with specific characteristics and tasks. In my research, I am focused on the individual and team-related aspects of boundary spanning and build on Kleinbaum and Tushman's (2007) definition, who defined boundary spanners as people who act as knowledge intermediaries among individuals within and outside the organization. Casciaro and colleagues (2019) found that diverse teams with people who promote cross-boundary work significantly outperformed teams without such people. However, Zhao and Anand (2013) argue that boundary spanning is only efficient for tangible, discrete knowledge and that more complex forms of collective knowledge require a set of inter-unit ties.

Nevertheless, the organizational behavior literature on boundary spanning has long recognized the informational advantages associated with access to external sources of knowledge (Tortoriello & Krackhardt, 2010), which should help teams to drive business and IT alignment that is necessary for sustainable team innovation success (Reynolds & Yetton, 2015). Such an information advantage resulting in better access to knowledge, experiences, and resources should amplify functional diversity's effects on team innovation. Drach-Zahavy and Somech (2001) support this conclusion and argue that factors contributing to informational resources are conducive to innovation.

Cummings (2004) argues that team performance increases when structurally diverse teams engage in external knowledge sharing, including but not limited to exchanging feedback with customers. Furthermore, Moser and colleagues (2019) presented evidence that teams were more innovative with greater occupational diversity at higher levels of information sharing and helping behavior in the team. This leads me to my first hypothesis:

Hypothesis 1. Functional diversity is positively related to team innovation, and this relationship is stronger for teams that practice boundary spanning.

Information Integration

As part of the knowledge integration theme, van Knippenberg (2017) put primary focus on information elaboration as the main process driving team innovation and making the case that the integration of diverse information and perspectives enables teams to be innovative; speaking to the very core reason of the team's existence and how it can be more innovative than any of the team members by themselves. Van Knippenberg (2017) proposed the mediating role of information elaboration to create synergies from informational resources and defined it as the team process of “*sharing, discussion, and integration of task-relevant information and perspectives.*” From a conceptual perspective, as shown in Figure 2, constructs that provide the team informational resources can be seen as inputs for the elaboration process, which is then driving team innovation. This is also evident in diversity research on team networks (e.g., Baer, 2010; Balkundi & Harrison, 2006) and team composition. Mathieu, Tannenbaum, Donsbach, and Alliger (2014) show that distributions of team member profiles facilitate team processes and effectiveness. This is not only consistent with findings from the diversity

literature but also points out that having access to informational resources does not mean that the team is naturally using them in the right way. Specifically, the task-coordination aspect of boundary spanning creates a focus on better alignment with customer needs through feedback loops and thus helps teams thrive with information elaboration. Research has shown that accessing knowledge from across boundaries is an imperative driver of team innovation (Ancona & Caldwell, 1992; Reagans & Zuckerman, 2001).

Further, in the context of my research, boundary spanning is explicitly operationalized as boundary spanning between the account team and the customer, which is intended to bring in information about the client and create a focus on the client's perspective to not only better understand the core issues at hand but also increase software usage to address these issues. This speaks to the broader issue in the software and technology industry where innovation is not just about idea generation or bringing products to market but to ensure implementation (West, 2002) and customer usage. Many companies directly work with clients to develop solutions specifically for the client. The business interest is that the client uses the solution and that this results in a continued relationship. In that sense, boundary-spanning can help to collaboratively develop a mutual understanding and address all aspects of the business relationship, including the strategic direction, business model and process alignment, solution functionality, IT organization and implementation partner alignment, risk mitigation, as well as services and support for ongoing operations. For fast value realization, boundary spanning is critical to bring in the customer's perspective to understand what users want and include the perspective of implementation partner(s) who typically have accountability for the initial adoption.

With a different research focus but similar context, Harvey, Johnson, Roloff, and Edmondson (2019) studied sales teams of a large financial services firm and explored how they coordinate their activities, share information about customers' needs and issues. They found that the relationship between team learning orientation and team learning was mediated by psychological safety, but only for teams with low open-mindedness. In that regard, it is also important to note that a common and widely used practice of innovation-focused account teams is design thinking, which enables the team to prioritize the user experience (Plattner, Meinel, & Leifer, 2011). For example, Seidel & Fixson's (2013) research outlines the advantages and limitations of design methods, explicitly demonstrating how high team reflexivity in the form of debating ideas, methods, and concept changes was associated with more successful outcomes. Although design thinking is essentially a form of boundary spanning including communication and collaboration across functions and with other companies, it is currently not linked the boundary spanning literature beyond foundational concepts, such as focus groups, prototyping, or brainstorming to generate ideas (Micheli, Wilner, Bhatti, Mura, & Beverland, 2019).

Design thinking is a design-led approach to human-centered innovation based on the notion that value can only be created if the needs of the people who use the innovation are understood clearly. It relates to the principles of experiential learning and is practiced as an iterative process of learning by doing, often including prototypes and early market tests of a minimum viable product (Cousins, 2018). Design thinking helps obtain a multi-perspective comprehension of complex and ambiguous problems and develop feasible, desirable, and economically viable solutions (Plattner et al., 2012).

Venkatesh, Digerfeldt-Mansson, Brunel, and Chen (2012) argue that design thinking is a recipe for innovation success. Gruber, de Leon, George, and Thompson (2015) further discuss using a design-process to innovate and focus on customer needs as a source of information. I argue that this is especially critical for complex services engagement where IT and business alignment is critical over time (Sabherwal et al., 2001). Therefore, by drawing on these notions, I propose:

Hypothesis 2. Functional diversity interacts with boundary spanning such that it is positively related to information elaboration for teams that practice boundary spanning.

Building on the learning literature, Tzabbar and Vestal (2015) elaborate that knowledge integration in a team requires not only access to unique information but also requires the team's ability to transform individual information into valuable innovation. Bunderson and Sutcliffe (2003) provided consistent evidence and found that an appropriate emphasis on learning improves team effectiveness. The main argument here, further building on van Knippenberg's (2017) propositions, is that knowledge exchange and integration are conducive to team innovation. This is consistent with the finding of Ancona and Caldwell (1992) that external communication plays a mediating role and demonstrates that the diversity of a team's external network can also be understood as an informational resource for the information elaboration process.

Somech and Khalaili (2014) found a positive relationship between inter-team goal interdependence, functional diversity, and team innovation that was mediated by scouting and coordinating boundary-spanning activities. Teams should be more innovative the more team members bring different perspectives and can create novel insight by engaging

in information elaboration to address innovation challenges. Drach-Zahavy and Somech (2001) presented supporting evidence for a positive relationship between functional diversity and team innovation mediated by team interaction processes that reflected the degree to which team members engaged in information exchange, learning, motivating, and negotiating. To enable team innovation success, customer engagement teams need to engage in boundary-spanning activities, such as exchanging information, know-how, and customer feedback (Cummings, 2004) as well as coordination (Drach-Zahavy & Somech, 2010; Marks, DeChurch, Mathieu, Panzer, & Alonso, 2005). In that regard, Faraj and Sproull (2000) also found a strong relationship between expertise coordination and team performance. Tortoriello and Krackhardt (2010) presented somewhat conflicting evidence. They showed that informational advantages of cross-organizational boundary spanning have no positive impact on innovative performance unless there are at least three boundary-spanning connections in a group. Zhang (2016) presented further evidence of a positive relationship between functional diversity and team creativity mediated by the two mechanisms of knowledge sharing and group cohesion but only for long-standing teams.

Mitchell and Boyle (2015) found that professional diversity enhances team innovation through professional identity salience with second-stage moderation through open-mindedness norms. Cheung, Gong, Wang, Zhou, and Shi (2016) presented conflicting evidence and found a negative effect of functional diversity on team innovation through knowledge sharing in teams with low trust and no significant relationship with higher trust. Miron-Spektor, Erez, and Naveh (2011) found that a mix of creative and conformist team members enhances a team's capability to drive radical

innovation, mediated by team potency. Somech and Drach-Zahavy (2007) found that functional diversity was positively related to team innovation mediated through team processes, such as information exchange and learning, that was also influenced by interaction frequency. Qu and Liu (2017) empirically verified a moderated mediation effect of informational diversity faultlines on team creativity through external knowledge acquisition and internal knowledge integration moderated by the team's focus on cooperation. Schotter and colleagues (2017) postulate that managing and coordinating across organizational boundaries emerged as a critical capability required for sustainable success. A coordinated and structured approach to innovation is needed to help teams with crucial activities, such as planning, prioritization, resource mobilization, change management, business / IT alignment, and feedback. Specifically, on the topic of feedback, the research of Hoever, Zhou, and van Knippenberg (2018) provides fascinating insights into how positive and negative feedback enhances creativity at the team level. They found a moderated mediation relationship between feedback valence and team creativity through information elaboration and generative processing with first stage moderation by informational diversity. Based on these arguments, I suggest the following hypothesis of a moderated mediation model where information elaboration mediates the relationship between functional diversity and team innovation, and boundary-spanning activities moderate the knowledge integration process.

Hypothesis 3. The positive relationship between functional diversity and team innovation through information elaboration is moderated by boundary spanning such that it is stronger for teams that practice boundary spanning.

Team Climate Perspective

The second major theme in team innovation research is the team climate perspective (Hülshager et al., 2009; West & Farr, 1990; van Knippenberg, 2017). Van Knippenberg and colleagues (2004) postulated that team diversity benefits are contingent on factors motivating information elaboration and factors preventing diversity's negative effects of social categorization. Schneider and Reichers (1983: 22) defined organizational climate as "*the shared perceptions of organizational policies, practices, and procedures.*" For innovation research, West (1990) shaped the team climate for innovation by defining it as the perception of whether innovation is valued and supported within the team. Anderson and West (1998) address the climate for work group innovation with a four-factor theory and developed a team climate inventory (TCI). They identified vision, participative safety, task orientation, and support for innovation as climate factors associated with team innovativeness. This logic suggests that innovative outcomes are driven by team members who share a commitment to common goals and excellence and believe that they can participate safely in team discussions contributing to a worthwhile innovation. Based on TCI, Kivimäki and Elovainio (1999) developed and validated a shorter 14-item version of the TCI and demonstrated an acceptable item coverage and predictive validity compared to the original version. Bain, Mann, and Pirola-Merlo (2001) found supporting evidence of a positive relationship between the climate for innovation and team innovation. The relationship between these factors and team innovation has also been confirmed by the meta-analysis of Hülshager and colleagues (2009). Specifically showing that the climate for innovation is positively related to team innovation in the sense that teamwork with a focus on shared goals is

important for team innovation. Lisak, Erez, Sui, and Lee (2016) found a positive influence of team-shared innovation goals. Peltokorpi and Hasu (2014), adding additional nuance, found that participative safety was more relevant for team innovation with increasing team size. In my research, I focus on the moderating effects of shared vision following the logic that a shared commitment to team objectives is positively related to team innovation. *“Vision is an idea of a valued outcome, which represents a higher-order goal and motivating force at work”* that provides the basis for action within the team (West, 1990: 310). Anderson and West (1998) summarize that for a team to commit to a shared vision, it requires clarity of the valued outcome, acceptance within the team, and attainability, meaning a relatively clear line of sight on how to achieve the desired innovation. Fay and colleagues (2006) support the argument and postulate that a shared vision implies the pursuit of appealing and worthwhile goals. The research from Pearce and Ensley (2004) is a good example in that regard showing not only the positive impact of a shared vision on innovation effectiveness but also demonstrating effectively that shared vision plays a key role in the team innovation process. By applying the same logic, Chen, Farh, Campbell-Bush, Wu, and Wu (2013) focused on support for innovation and found a positive relationship with team innovation. Eisenbeiss, van Knippenberg, and Boerner (2008) further studied interactions between components of climate for innovation and found, besides important insights on transformational leadership, that team innovation only benefited from support for innovation in teams that had a climate for excellence. In yet another spin, Lyubovnikova, West, Dawson, and West (2018) found a positive relationship between shared objectives and team productivity and innovation by exploring the mediating process of generating shared

objectives. Gong, Kim, Lee, and Zhu (2013) furthermore found that learning goal orientation is positively associated with team innovation. In summary, teams are more likely to successfully implement ideas when they have a shared vision giving them focus and direction and allowing them to develop appropriate methods and practical steps. Supporting this notion, Fay and colleagues (2006) found a positive interaction effect of team climate and diversity predicting team innovation. More specifically, the quality of team processes, operationalized via a shared vision, a safe climate, working closely, and taking a reflexive approach to work, moderated the relationship between team diversity and innovation such that innovation quality benefited from increasing levels of diversity only in the case of good team processes. As a result, the following hypothesis is suggested:

Hypothesis 4. Functional diversity is positively related to team innovation, and this relationship is stronger for teams that share a common vision.

Further building on van Knippenberg (2017) and team learning theories that emphasize the importance of social structures required for effective communication and knowledge sharing (Argote, McEvily, & Reagans, 2003; Tzabbar & Vestal, 2015), it is argued that the team climate for innovation moderates the relationship between informational resources and information elaboration. In that sense, the two main themes of knowledge integration and team climate complement each other. The team climate comprises factors that motivate the team to use informational resources in the information elaboration process to drive innovation. By differentiating idea generation and innovation, Somech and Drach-Zahavy (2013) found evidence for a moderated mediation model showing that both functional background diversity and creativity of the team

members predicted innovation through team creativity, but only for teams that operated in a favorable climate for innovation. This provides further evidence that the positive effects of informational resources are moderated by factors encouraging and stimulating their use.

Similarly, the research from van Ginkel and van Knippenberg (2008) suggests that the team climate for innovation has a positive moderating influence on the relationship between team informational resources and information elaboration. If team members can speak up and take risks, they can generate ideas and identify alternative problem-solving approaches when facing barriers during the implementation. In contrast, if they do not feel they can contribute freely and speak their mind, then diversity has no positive effect on elaboration (Edmondson & Lei, 2014). Rousseau, Aubé, and Tremblay (2013), also looking at antecedents of the team climate, found that goal commitment and support for innovation were positively related to team innovation, and both were predicted by team coaching. This leads to my fifth hypothesis, framing these insights for a shared vision and arguing that shared vision motivates integrating different functional perspectives in the team information elaboration process.

Hypothesis 5. Functional diversity interacts with shared vision such that it is positively related to information elaboration for teams that share a common vision.

In further building out the knowledge integration / team climate model of team innovation, like in hypothesis 3, I argue that team innovation benefits from functional background diversity when teams engage in information elaboration and operate in a positive climate for innovation (van Knippenberg, 2017). In that regard, Hoever and colleagues' (2012) research is a good example that provides experimental evidence for a

moderated mediation effect of diversity of perspectives on team creativity through information elaboration that is conditional on member perspective-taking.

Further extending team creativity research, Bodla, Tang, Jiang, and Tian (2018) presented in their study evidence that an inclusive climate moderates the diversity-creativity relationship, mediated by knowledge sharing. They also provided specific recommendations for managers to cultivate an inclusive climate by appreciating difference, providing equal opportunities, and involving team members in decision making as a highly inclusive climate will mitigate the adverse effects of demographic diversity.

In related IT research, Sharma, Mihas, and Kankanhalli (2014) explored how organizations can create value from big data and business analytics. They argue that managers need to transform their decision-making and resource allocation processes to capture the value of business analytics. Kukenberger and D'Innocenzo (2020) provided further evidence for moderation effects showing that, on the one side, shared leadership benefits from functional diversity in teams operating in a cooperative climate. On the other side, demographic diversity harmed shared leadership for teams working in a less cooperative climate. Resick and colleagues (2014), in a similar direction but further exploring second-stage moderation effects, showed that teams benefit from information elaboration, specifically in turbulent environments. More formally, I hypothesize the following:

Hypothesis 6. The positive relationship between functional diversity and team innovation through information elaboration is moderated by shared vision such that it is stronger for teams that share a common vision.

The above-presented research studies are consistent and well-aligned with the knowledge integration and team climate perspective (van Knippenberg, 2017) and team diversity theory (van Knippenberg et al., 2004). The discussed evidence and findings show significant relationships between functional diversity and team innovation with mediation through information elaboration or similar constructs and moderation by team climate and other factors. Depending on the moderating conditions, diversity can have positive or negative effects on information elaboration. Specifically, if teams do not operate in a favorable climate for innovation and lack motivation for teamwork, then social categorization processes can prevail, leading to communication breakdowns, disruption, and conflict. Whereas, if team members are motivated to integrate diverse perspectives, diversity is a positive force driving team innovation.

Research Model

The integrated conditional process model of my research is shown in Figure 3. While its heritage from the knowledge integration / team climate model of team innovation is unmistakably clear regarding the dependent and mediating variables, the independent and moderating variables were more precisely defined. Measures and scales for each construct are specified in the measures section of the following methods chapter.

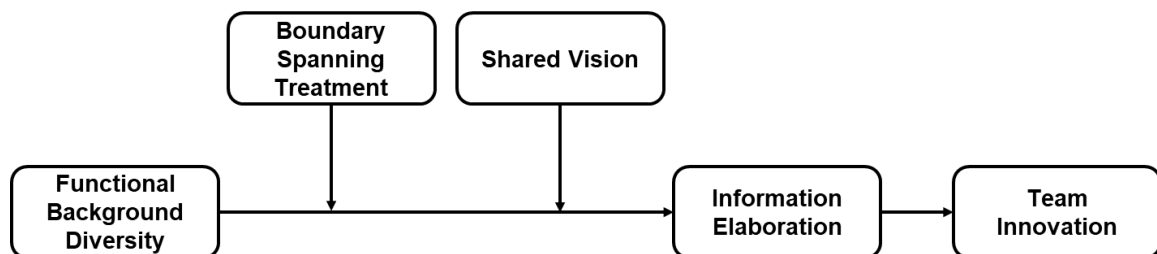


Figure 3 Research Model

In summary, I conclude this chapter with the main arguments inherent in my research model. Factors contributing to the team's informational resources are conducive to innovation. Building on the extensive team diversity literature, I explore the effects of functional background diversity. Teams are more innovative, the more team members bring different functional perspectives and engage in information elaboration. Information elaboration is the critical mediating team process of knowledge integration to address innovation challenges. Positive effects of functional diversity come through information elaboration and are contingent on factors that stimulate the integration of different perspectives. Boundary spanning and a shared commitment to common team goals are considered as such factors. Therefore, cross-functional teams with high shared vision are more likely to drive team innovation when engaging in information elaboration successfully. Likewise, teams that practice boundary spanning, such as scouting and task-coordination, will better integrate the different perspectives resulting in more creative ideas and targeted problem solving, ultimately more team innovation success.

CHAPTER 3: METHODS

Research Design

Since research on team innovation is maturing in terms of operationalization of constructs, integration of perspectives and theories, as well as the availability of well-grounded building blocks (van Knippenberg, 2017), choosing a quantitative research approach appeared to be appropriate and allows for better generalization of results (Shadish, Cook, & Campbell, 2002: 348). I utilized a cross-sectional company survey (positivist, quantitative approach) to gather information about the shared vision of team members, information elaboration team processes, and validate team member functional backgrounds along with certain control variables, such as team tenure and executive support, that are explained later on in this section together with all other measures. I developed a new survey instrument for the data collection by reusing empirically validated measurement scales for the specified constructs (see Appendix A).

To further establish causal relationships for the impact of boundary spanning, I obtained information from the company's field experiment and combined it with the survey results. The experiment was conducted from January through July 2020 and involved 162 customer engagement teams working actively with North American customers across industries. I used the experiment team innovation data from January as baseline and July data as a result. The survey was deployed from May 26 through July and was closed before the July team innovation outcome measurement on August 7, 2020. With the combined survey and experiment data, I was able to test the boundary-spanning assertions and their impact on team innovation through a quasi-experimental intervention that introduced an additional team role charged with boundary-spanning

practices. The main tasks of the boundary-spanning role were coordinating resources across departments, focusing on the customer’s business outcomes, relationship management, ongoing process governance, and communication. For further details about the intervention, please refer to the measures section. As a result, the company expected that teams that received the treatment demonstrate more successful team innovation—facilitated by a better understanding of the product, targeted customer onboarding, end-user training, and results-oriented solution adoption. Further, in the sense of overall better relationships with customers, it was expected that as a result, customers are more likely to accept and implement their solutions in full. The research and data collection timeline are illustrated in Figure 4.

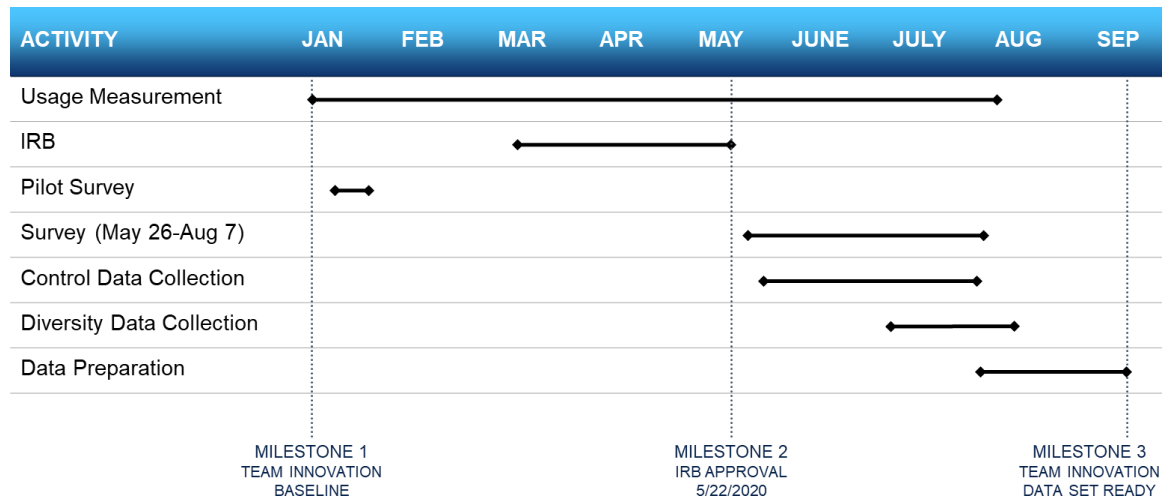


Figure 4 Research and Data Collection Timeline

Data and Sample

The unit of analysis is the customer engagement team with team sizes ranging from 3 to 27 and a median of 9 members, including but not limited to sales reps, industry and value advisors, solution and support engineers, enterprise architects, engagement

managers, as well as product and adoption experts. In the company experiment, teams were randomly assigned to either the treatment or control group. Teams that were part of the treatment group received an additional team member charged with boundary-spanning activities. In 48% of the teams with treatment status, an existing team member was promoted for the new role, and his or her old position was backfilled. This is consistent with a recommendation of Richter, West, van Dick, and Dawson (2006) and Tushman and Scanlan (1981). They argue that the promotion of technically competent employees with strong connections to their team and clients helps overcome dysfunctional relationships. In 20% of the cases, boundary spanners served multiple customer engagement teams. A typical treatment team is illustrated in Figure 5.

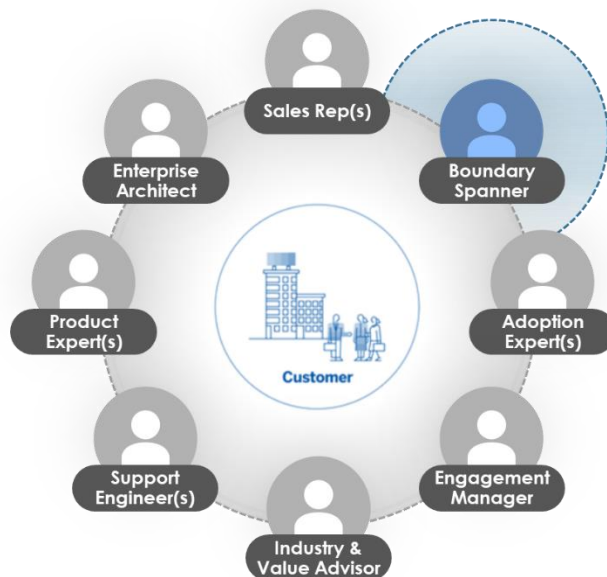


Figure 5 Composition of a Customer Engagement Team

The teams in the treatment group had to obtain consent from their customers for program participation. Five customers rejected the treatment. I did not receive any survey responses for one of the five customers and therefore excluded that team from the

analysis. For another customer, it turned out that this company was merging with a customer that was also in the treatment group and already had a boundary spanner assigned. One customer initially rejected the treatment because of an ongoing restructuring but later agreed to participate. Finally, two customers chose not to participate, and therefore I excluded them from the analysis (see also footnote 1 in the results chapter).

Although the experiment customers represented only about 1% of the entire customer base by count, they represented 36% of the total revenue. They thus can be considered representative of the mid- to the top-segment business of the company. As shown in Figure 6, the customers in the experiment furthermore represented the right mix of industries.

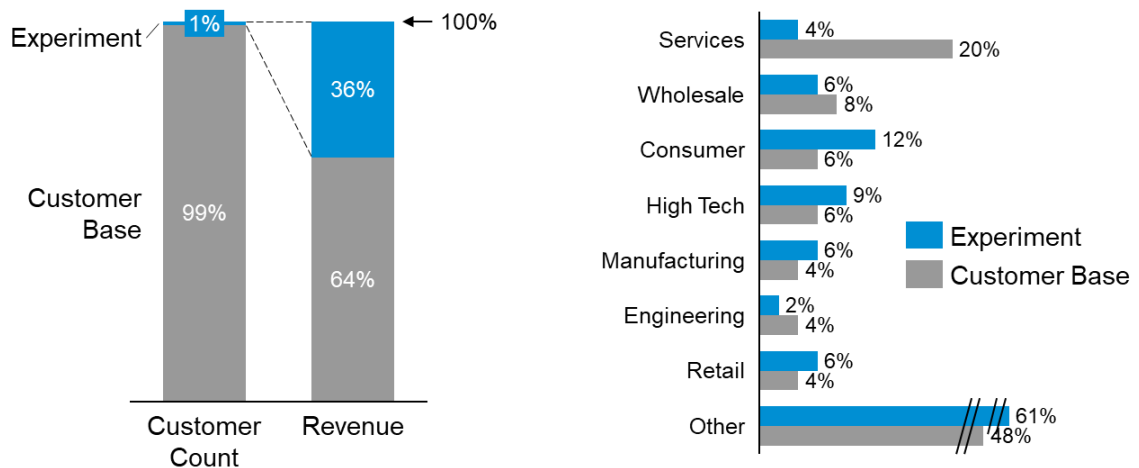


Figure 6 Experiment Sample Population in Context of the Entire Customer Base

The developed survey instrument was piloted individually with five subject matter experts. Based on the feedback, I made some minor wording changes such that participants better understand that the questions reference the team innovation

environment and related mechanisms. The survey participation was voluntary, and employees who were not interested in completing the survey could ignore the survey email. I sought responses from the company employees who had worked actively in a customer engagement team for at least six months and gained experience in a customer-facing capacity from the experiment participation in either the treatment or control group. While many participants were based at the company headquarter, it was not required to be onsite to participate in the study. The survey was constructed and made available via Qualtrics, an online web application, and participants who decided to take the survey could complete it at a time and location of their choice. The survey link was sent via email to 1,519 employees. Employees with back-office functions or engagements with customers in other geographies, such as Europe or Asia, were not eligible to participate in this study and were not contacted. Vulnerable populations, such as adults unable to consent, children, or prisoners, were not included in the sample. Pregnant women, however, were not excluded. I did not collect pregnancy information, and therefore I am not aware whether a participant was pregnant.

In order to be able to aggregate individual team member ratings to the team level, the respondents were asked to select their customer team. A default value was provided based on company records for each respondent to simplify the team's selection from a drop-down list with over 160 entries. Since respondents could be part of multiple teams, they could change the default value. The respondents were asked to consider answering the survey questions for their most relevant customer engagement to get the best quality insight on team innovation. In order to provide the accurate team default, the Qualtrics survey distribution capabilities with personalized links were leveraged. With the help of

JavaScript code, the default customer team was set for each respondent. It took respondents between 10 to 15 minutes to complete the survey. After the data collection, all data were deidentified. According to the Institutional Research Board protocol, personally identifiable information in individual survey responses was split-off and kept strictly in separate files. The information is linked via a response identifier if there should arise a need to trace back to an individual respondent. There were no anticipated risks and inconveniences beyond filling out the survey.

Further, participants who elected to be part of the study could skip any items that they did not feel comfortable answering or terminate participation in the study at any time. There were also no direct benefits for participating in the study. Because of no more than minimal risk of harm to subjects, no written documentation of consent was required. The Institutional Research Board of the Drexel University approved the HRP-501 research protocol “Exploring the knowledge integration and team climate perspectives for team innovation” (protocol number 2004007759) on May 22, 2020.

Measures

Functional Background Diversity

Based on my theoretical approach and in alignment with previous research (e.g., Kukenberger & D'Innocenzo, 2020; Zhang, 2016), I operationalized the diversity of team member functional backgrounds with Blau's index of heterogeneity normalized for varying team sizes as recommended by Biemann and Kearney (2010: 585) across the following eleven categories of business functions: Distribution & Logistics, Finance & Accounting, Human Resources, Information Technology, Marketing & Communications, Production, Research & Development, Supply Chain & Operations, Sales & Customer

Service, Sourcing & Purchasing, Strategy & Business Development. I calculated the diversity index for each team based on available survey responses and company records for team members that did not submit their survey. The index ranges from zero indicating no functional diversity to one indicating maximum diversity of functional backgrounds among team members.

Boundary Spanning Treatment

Teams with treatment status received an additional team member that was charged with boundary-spanning activities. These activities include ambassadorial activities to persuade stakeholders to support the team's goals, access and mobilize critical resources. Scouting activities to gather relevant customer and market information and task-coordinator activities to orchestrate key team processes focusing on the customer (Ancona & Caldwell, 1992). The boundary spanner's main tasks were coordinating resources across departments, focusing on the customer's business outcomes, gathering information, ongoing process governance, relationship management, and communication. In other words, the overall orchestration of the implementation and adoption process by understanding the fundamental problems to be solved, planning the best solution with empathy for the customer, and creating an inclusive team climate that involves everyone's perspective. The boundary-spanning team members were trained for this specific customer engagement methodology before introducing them to the team and their customers. Upon the beginning of their engagement, the boundary spanners were tasked to organize a customer relationship assessment and with it reset the collaboration no matter in which engagement phase the customer was at the time (for example, exploration of new solutions or adoption of already acquired solutions). With the training,

the company ensured that all boundary spanners operated under the same set of guidelines and operating parameters. It is also noteworthy that the company did a pilot test of the boundary-spanning intervention with 30 customers in 2019, which resulted in several refinements of the engagement methodology and the training materials that were later utilized in the 2020 field experiment.

The boundary-spanning treatment was measured with a binary variable separating treatment and control group wherein zero identified the control group and one the treatment group.

Shared Vision

I adapted six items from Anderson and West (1998: 241) as defined in their team climate inventory to measure shared vision, for example, “How clear are you about what your team’s objectives are?” and “To what extent do you think they are useful and appropriate objectives?” Team members were asked to rate each item for their team on a 7-point scale ranging from “not at all” to “completely.” Cronbach’s alpha of .95 indicated excellent internal homogeneity and reliability. The calculated agreement index median R_{wg} of .87 and intraclass correlations $ICC_{(1)}$ of .74 and $ICC_{(k)}$ of .95 justified aggregation to the team level (LeBreton & Senter, 2008).

Information Elaboration

To assess information elaboration, I reused the scale of van Knippenberg, van Ginkel, Stam, Giessner, van Balen, and Ghorbani (2019). Sample items include “In our team members complement each other by sharing their knowledge” and “We discuss alternatives before making decisions.” As part of the survey, respondents were asked to indicate their agreement to these items within their team on a 5-point scale ranging from

“strongly disagree” to “strongly agree.” Cronbach’s alpha of .96 indicated excellent internal homogeneity and reliability. The calculated agreement index median R_{wg} of .87 and intraclass correlations $ICC_{(1)}$ of .80 and $ICC_{(k)}$ of .96 justified aggregation to the team level (LeBreton & Senter, 2008).

Team Innovation

Team innovation was measured as the usage of an enterprise software technology solution within the adopting customer organization. The utilization of the implemented technology solution was represented as a percentage of the actual metered usage over the purchased software entitlement. For example, if a customer purchased 500 users of a particular solution and 100 users actually used it, the team innovation index would be 20 percent. The utilization percentages were calculated for all participating customers with an analytics report of the company. They were made available for the beginning (baseline) and at the end of the measurement period. The index ranges from zero indicating no use to one indicating full use. In case a customer used multiple software solutions integrated into one business process, then the team innovation index was calculated as the mean across all solutions.

Controls

I included the following control variables in my analyses to partial out their effects on team innovation: Team size, executive support, team tenure, and experience, as well as several customer attributes, such as industry, business familiarity, annual revenue, and COVID-19 pandemic impact.

Team size was measured as the sum of core team members. The company separates between the core and extended team. By contrast, members of the extended

team only join the engagement on a temporary or as-needed basis and do not, for example, participate in regular team meetings. Accordingly, I focused on the core team in my analysis. I included team size as a control variable because it is often found in team research that it influences team performance (Fay et al., 2006; Hülshager et al., 2009; Moser et al., 2019). Research evidence suggests that smaller and larger teams differ in many important ways. While larger teams can enjoy access to more resources, higher levels of diversity, and may appear more legitimate, they can also experience more coordination and communication challenges, member task conflict, and less cohesion (Mathieu, Maynard, Rapp, & Gilson, 2008; Moreland, Levine, & Wingert, 1996 in Witte & Davis, 2014).

Executive sponsorship and top management support have been identified as critical success factors for innovation with information systems (e.g., Chandler & Thomas, 2015; Lucas et al., 2007; Shao, Feng, & Hu, 2017; Sharma & Yetton, 2003), specifically because of resources intensive implementation projects and the ongoing need for end-user support and training. Executive sponsors are typically important not only during the initial stages of a customer engagement to ensure alignment and help mobilize required resources but are also critical for ongoing governance, business reviews, conflict resolution, and prioritization (Drucker, 2002). Furthermore, van Meerkerk and Edelenbos (2017) found an indirect positive relationship between executive support and boundary spanning. Therefore, I also controlled for differences from executive support. I have asked team members to rate how pleased they are with the executive sponsorship of their customer engagement using a 5-point scale ranging from “extremely displeased” to “extremely pleased.”

Team tenure was also included as a control variable since it has been identified as a critical component in team effectiveness models. Gonzalez-Mulé, Cockburn, McCormick, and Zhao (2019), for example, found that team tenure was positively related to team performance. Zhang (2016) furthermore found that team longevity positively moderated the relationship between functional diversity and team creativity through knowledge sharing and group cohesion.

Following the group development theory of Kozlowski, Gully, Nason, and Smith (1999), I calculated the collective team tenure as the mean amount of time individual members have been together on the same team, allowing them to gather shared experiences. The data has been collected with the survey instrument wherein I asked team members how long they worked with their team. Beyond the collective tenure, building on Becker's (1964 in Gonzalez-Mulé et al., 2019: 154) human capital theory, I measured the work experience of the team, which is also referred to as additive team tenure capturing the differences in skills, experience, and knowledge available within the team. This is consistent with team composition research (e.g., Bell, 2007), where individual attributes of team members are aggregated to team-level factors. Work experience has been captured with a simple survey question asking respondents for how many years they have been in the workforce. Like Haas (2010), I calculated it as the mean amount of years that team members have spent in the workforce.

Finally, I controlled for the industry segment of the customers. The industry data for each team and customer has been obtained from company records. Customers were assigned to one of the following six industry clusters: Consumer, manufacturing, energy & natural resources, financial services, public sector, and services industries.

The industry cluster assignment for each team was coded with dummy variables. Besides the industry, I also controlled for business familiarity measured as the number of days since becoming a customer, its annual revenue, and COVID-19 impact. Specifically, for customers impacted by the COVID-19 pandemic, the company data set included an indicator flagging customers who requested concessions, such as extended payment terms or deferred maintenance payments, based on hardship situations in their industry, most prominently in the airline or hospitality industry.

Manipulation Checks

In order to assess the effectiveness of the experimental intervention, I used the following direct and indirect manipulation checks (Ejelöv & Luke, 2020). First, I checked directly whether a boundary-spanning team member was assigned to the treatment group teams. This information was available in the data set of the company experiment. Beyond this essential check, I analyzed whether the boundary spanners performed the trained intervention. As elaborated above, boundary spanners were trained to start or reset the customer engagement with a relationship assessment workshop and document goals and progress in an outcome success plan. In my survey, I asked team members to what extent they agree or disagree (5-point scale ranging from “strongly disagree” to “strongly agree”) that an outcome success plan was developed by the team and shared with the customer since this represents the most salient and tangible measure of the intervention.

Furthermore, I reused Faraj and Yan’s (2009: 611) scale to measure boundary spanning team activities with my survey instrument. Respondents were asked to indicate their agreement to the four items on a 5-point scale ranging from “strongly disagree” to “strongly agree.” For example: “Does your account team value team members for making

use of their relationships with others on behalf of the team?” Cronbach’s alpha of .91 indicated excellent internal homogeneity and reliability. The calculated agreement index median R_{wg} of .79 and intraclass correlations $ICC_{(1)}$ of .70 and $ICC_{(k)}$ of .90 justified aggregation to the team level (LeBreton & Senter, 2008).

Analytical Approach

To test the hypothesized model of relationships, I utilized the IBM SPSS software version 26 with the PROCESS macro version 3.4 developed by Hayes (2018a). PROCESS uses an ordinary least squares or logistic regression-based path analysis to estimate direct and indirect effects in mediator models, moderation models, and conditional effects in moderated mediation models (Hayes, 2018a).

Before testing the hypothesized relationships, I first tested scale reliability and validity, calculated R_{wg} as an assessment of the within-group agreement and intraclass correlations (ICCs) to ensure the appropriateness of item aggregation (LeBreton & Senter, 2008). All R_{wg} values did reach the defined cut-off values of .7 (Klein et al., 2000). I also assessed how well the previously empirically validated measurement items represented the constructs with a confirmatory factor analysis. According to Kline (2005), I used the following suggested cut-off values: Comparative fit index (CFI) $\geq .9$, non-normed fit index (NNFI) $\geq .95$, root mean square error of approximation (RMSEA) $< .08$, and a non-significant model chi-square p-value $> .05$. The expected three-factor model demonstrated good model fit ($X^2(101) = 550.03$, $p < .01$, CFI = .97, NNFI = .95, RMSEA = .07). I then aggregated individual survey responses to the team level and computed the constructs from the single items. Afterward, I merged the team survey data with the team diversity and team innovation experiment data.

I mean centered non-binary study and control variables (Cohen, Cohen, West, & Aiken, 2003). For creating and testing interaction terms, I relied on PROCESS since it automatically does that for the modeled interactions (Hayes, 2018a).

I evaluated the fit of the model and the statistical significance of the interaction terms as well as direct and indirect effects to determine if the hypotheses were supported (or, more precisely, if the null hypotheses could be rejected). In doing so, I utilized the PROCESS Model 10, which is designed to estimate, test, and probe interactions as I have specified in my research model. It is important to note that in PROCESS Model 10, functional diversity is an additive linear function of boundary spanning and shared vision (Hayes, 2018b), meaning that the change in functional diversity's effect as boundary spanning changes is independent of the value of shared vision. Likewise, the change in functional diversity's effect as shared vision changes is independent of boundary spanning. The moderated mediation test is based on an interval estimate of the parameter that links the indirect effect to moderator values, providing moderated mediation indices (Hayes, 2018a).

I used the setting of 20,000 iterations for the bootstrapping analyses. Bootstrapping is considered the right approach for studies of indirect effects with small sample sizes (MacKinnon, Lockwood, & Williams, 2004).

CHAPTER 4: RESULTS

Descriptive Statistics

A total of 935 team members completed the survey, yielding a response rate of 62%. Thirty-nine responses (3%) did not have a team specified and were not included in the analysis. Individual team member responses were aggregated to the team level by computing the means resulting in 162 team cases. Sixteen teams with only one response and two treatment teams where the customer did not consent¹ were filtered out and not included in the analysis. The remaining 144 teams are the basis for the analysis, of which 77 received the treatment, and 67 constituted the control group. These 144 teams did meet the sampling ratio cut-off value of 0.17 (Dawson, 2003; Richter et al., 2006) and had response rates of at least 60%. The average team size was 9 ($SD = 4.33$). Table 1 reports the descriptive statistics and correlations for all study variables.

Table 1 Means, Standard Deviations, and Correlations

Variable	Mean	S.D.	1	2	3	4	5	6	7	8	9
1. Team Innovation	50.08	26.78									
2. Information Elaboration	4.16	0.57	.12								
3. Shared Vision	5.56	0.66	-.01	.29**							
4. Functional Diversity	0.79	0.17	.22**	.51**	-.04						
5. Boundary Spanning	0.53	0.50	.21**	.38**	.18*	.37**					
6. Innovation Baseline	44.53	27.25	.78**	-.20*	-.15	-.08	-.02				
7. Team Size	8.89	4.36	-.04	.10	-.04	.26**	.30**	.00			
8. Executive Support	3.56	0.65	-.15	.04	.39**	-.08	.14	-.18*	-.01		
9. Team Tenure	1.78	0.41	.06	.16	.17*	.10	-.18*	.08	-.02	.04	
10. Work Experience	23.11	5.39	.07	.25**	.01	.29**	.14	-.05	.10	-.05	-.01

Notes. N=144. * $p < .05$. ** $p < .01$

¹ In order to make sure that I do not bias my results, I also ran the analysis with these two teams included in the treatment group, maintaining group integrity. Alternatively, I included these teams in the control group to check that I do not underestimate the treatment effect. Either way, my results did not change notably. An intention-to-treat analysis was not justified because of >97% compliance.

Team innovation was positively related to functional diversity ($r = .22, p < .01$) and boundary spanning ($r = .21, p < .01$), and highly correlated with the team innovation baseline ($r = .78, p < .01$). It is also important to note that boundary spanning is correlated with functional diversity ($r = .37, p < .01$). While the boundary spanner as a full team member is included in the functional diversity measure, it is not the cause for that correlation. Even when excluding the additional boundary-spanning role from the team functional diversity calculation, boundary spanning is still correlated to functional diversity ($r = .32, p < .01$). The following figure shows how the analyzed teams were distributed across customer engagement phases. 53% of the teams worked on implementing and adopting the acquired software solutions, and 47% focused on ideation and exploration of new opportunities.

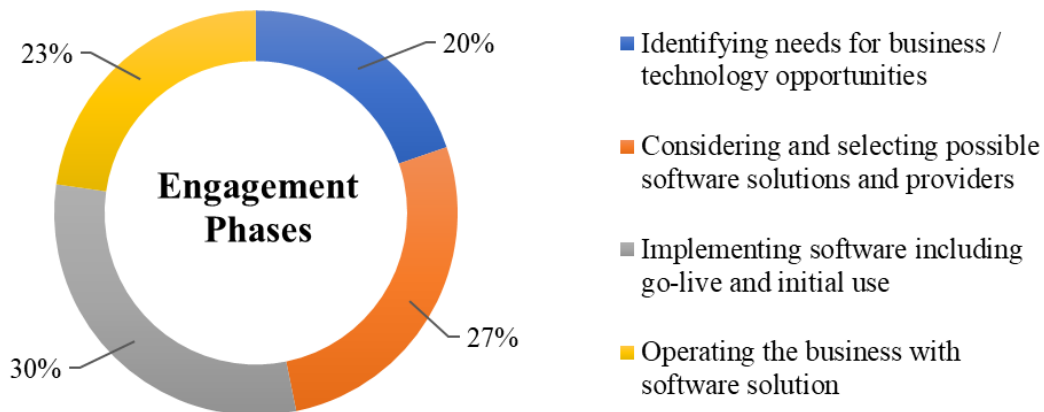


Figure 7 Team Distribution Across Engagement Phases

66% of the customer engagements were classified as incremental innovation to optimize the customers' business by team members. Examples include implementing recruiting and talent management solutions, replacing a legacy enterprise resource planning system, and exploring new solutions to make supply chains more resilient.

28% of the customer engagements were classified as disruptive innovations with a transformational character for the customer's business. Team members specified examples that included the use of predictive analytics solutions to automate procurement and finance functions, database solutions for digital platform business models, and expansion into new industry and market segments. The remaining 6% of engagements were described as a combination of both.

Across all team members, 75% were male and 25% female. 54% of the survey respondents have a bachelor's and 41% a master's degree. The average work experience was 23.5 years ($SD = 9.31$). The functional backgrounds of team members were distributed as shown in Figure 8.

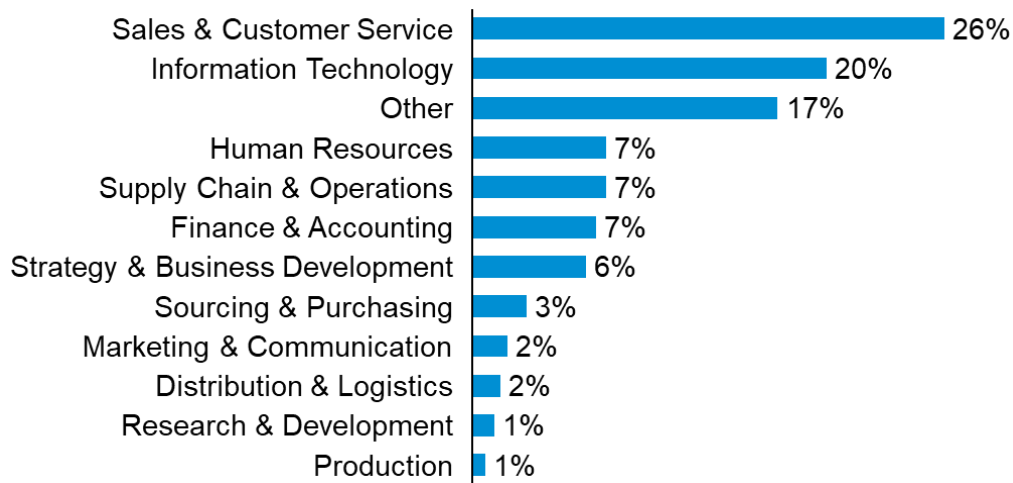


Figure 8 Distribution of Functional Backgrounds Across Team Members

The most common functional backgrounds were sales and customer services (26%) and information technology (20%). 17% of the team members described their function as “other” and provided detailed backgrounds, such as supply chain, sales and distribution, logistics, and production in the aerospace industry.

Finally, before going into the data analysis, I will assess and report the effectiveness of the manipulations (Ejelöv & Luke, 2020). Independent *t*-test analysis of the teams' boundary-spanning practices (Faraj & Yan, 2009) shows a statistically significant difference between the treatment and control group, providing evidence that the intervention has been effective. The 77 teams in the treatment group ($M = 4.05$, $SD = .40$) compared to the 67 teams in the control group ($M = 3.78$, $SD = .60$) demonstrated significantly higher levels of boundary spanning, $t(142) = 3.21$, $p < .01$. The teams in the treatment group ($M = 11.03$, $SD = 6.65$), furthermore, reported significantly higher levels of outcome success planning than the control group ($M = 6.74$, $SD = 3.85$), $t(129) = 4.36$, $p < .01$. This provides further evidence that the boundary-spanning team members performed the trained interventions.

In contrast to the boundary-spanning manipulation, the teams in the treatment group did not significantly vary on executive support and teamwork experience compared to the control group. There was no significant effect for executive support, $t(142) = -1.65$, $p = .10$, despite the treatment group ($M = 3.65$, $SD = .56$) reported slightly higher scores than the control group ($M = 3.47$, $SD = .73$). Similarly, there was no significant difference in teamwork experience, $t(142) = -1.72$, $p = .09$, for the treatment group ($M = 23.82$, $SD = 4.10$) compared to the control group ($M = 22.28$, $SD = 6.50$). Furthermore, individuals in the treatment and control groups did not significantly vary on demographic variables. Team members in the treatment ($M = 1.28$, $SD = .45$) and control groups ($M = 1.26$, $SD = .44$) were not significantly different on gender, $t(654) = -.56$, $p = .57$. There was also no significant difference in education between the members in the control ($M = 3.39$, $SD = .58$) and treatment groups ($M = 3.37$, $SD = .57$), $t(659) = -.36$, $p = .72$.

Data Analysis

Table 2 reports the regression results and PROCESS outputs for my research model that suggests a moderated mediation effect of functional diversity on team innovation through information elaboration that is conditional on boundary spanning and shared vision.

Table 2 Direct and Indirect Effects of Functional Diversity on Team Innovation

	Info. Elaboration			Team Innovation		
	Coeff.	SE	p	Coeff.	SE	p
Constant	.14	.34	.68	-.51	.27	.07
Functional Diversity	.22	.08	.01	.13	.07	.05
Boundary Spanning Treatment	.41	.13	<.01	.19	.11	.08
Functional Diversity x Treatment	-.34	.16	.03	.27	.13	.03
Shared Vision	.48	.07	<.01	.03	.07	.61
Functional Diversity x Shared Vision	.36	.05	<.01	.03	.04	.44
Team Innovation Baseline	-.11	.06	.07	.85	.05	<.01
Information Elaboration	—	—	—	.14	.07	.05
Team Size	-.05	.07	.49	-.09	.05	.08
Executive Support	-.09	.06	.16	-.01	.05	.86
Team Tenure	.07	.06	.24	-.02	.05	.69
Work Experience	-.05	.06	.46	.04	.05	.42
Consumer Industries	-.27	.36	.45	.37	.28	.20
Services Industries	-.27	.39	.48	.34	.31	.26
Manufacturing Industries	-.35	.36	.32	.38	.28	.18
Energy & Resources Industries	-.36	.37	.33	.44	.29	.13
Financial Industries	-.08	.39	.85	.09	.31	.77
Business Familiarity	.04	.06	.47	-.06	.05	.21
Annual Revenue	.02	.06	.72	-.06	.05	.20
COVID-19 Impact	-.06	.26	.83	-.18	.21	.38
	$R^2 = .64$			$R^2 = .78$		
	$F(18,125) = 12.36,$			$F(19,124) = 22.58,$		
	$p < .01$			$p < .01$		

Notes. N = 144. Values are standardized coefficients.

With PROCESS I first regressed information elaboration on functional diversity ($\beta = .22$, $SE = .08$, $p < .01$), boundary spanning ($\beta = .41$, $SE = .13$, $p < .01$), shared vision ($\beta = .48$, $SE = .07$, $p < .01$), and the interaction terms of functional diversity and boundary spanning ($\beta = -.34$, $SE = .16$, $p < .05$) and functional diversity and shared vision ($\beta = .36$, $SE = .05$, $p < .01$) as well as the team innovation baseline ($\beta = -.11$, $SE = .06$, ns) and the control variables, which yielded a significant R^2 of 64%. 26% of R^2 change was explained by functional diversity ($F(1,142) = 48.84$, $p < .01$), 33% by shared vision and its interaction with functional diversity ($F(2,140) = 56.07$, $p < .01$). Boundary spanning and its interaction with functional diversity explained 3% R^2 change ($F(2,138) = 5.14$, $p < .01$), and 2% were explained by the controls ($F(13,125) = .67$, ns). Next, team innovation was regressed on information elaboration ($\beta = .14$, $SE = .07$, $p < .05$), functional diversity ($\beta = .13$, $SE = .07$, $p < .05$), boundary spanning ($\beta = .19$, $SE = .11$, ns) and its functional diversity interaction ($\beta = .27$, $SE = .13$, $p < .05$), shared vision ($\beta = .03$, $SE = .07$, ns) and its functional diversity interaction ($\beta = .03$, $SE = .04$, ns), as well as the team innovation baseline ($\beta = .85$, $SE = .05$, $p < .01$) and the control variables, which yielded a significant R^2 of 78%. The biggest portion of 61% R^2 change was not surprisingly explained by the team innovation baseline ($F(1,142) = 221.43$, $p < .01$). 5% R^2 change was explained by the treatment ($F(1,141) = 22.08$, $p < .01$), 4% by information elaboration ($F(1,140) = 20.51$, $p < .01$), 2% by functional diversity² ($F(1,139) = 10.28$, $p < .01$), and 2% by the interaction of treatment and functional diversity ($F(1,138) = 7.65$, $p < .01$). The controls added another 4% R^2 change ($F(14,124) = 1.41$, ns).

² The functional diversity measure includes the boundary-spanning role. The conclusions are the same when the boundary spanner is excluded from the analysis. I still found a moderated mediation effect and there were no notable changes on information elaboration. The diversity treatment interaction and information elaboration however, exceeded the .05 significance threshold for team innovation.

As illustrated in Figure 9, to better discern the nature of the significant interaction of functional diversity and shared vision, I examined a simple slope computation at “low” (1 *SD* below the mean; $\beta = -.31$, $SE = .11$, $p < .01$) and “high” (1 *SD* above the mean; $\beta = .40$, $SE = .08$, $p < .01$) levels of shared vision (Aiken & West, 1991). These results indicate that functional diversity is positively related to information elaboration only for teams with a high shared vision. Therefore, Hypothesis 5 is supported.

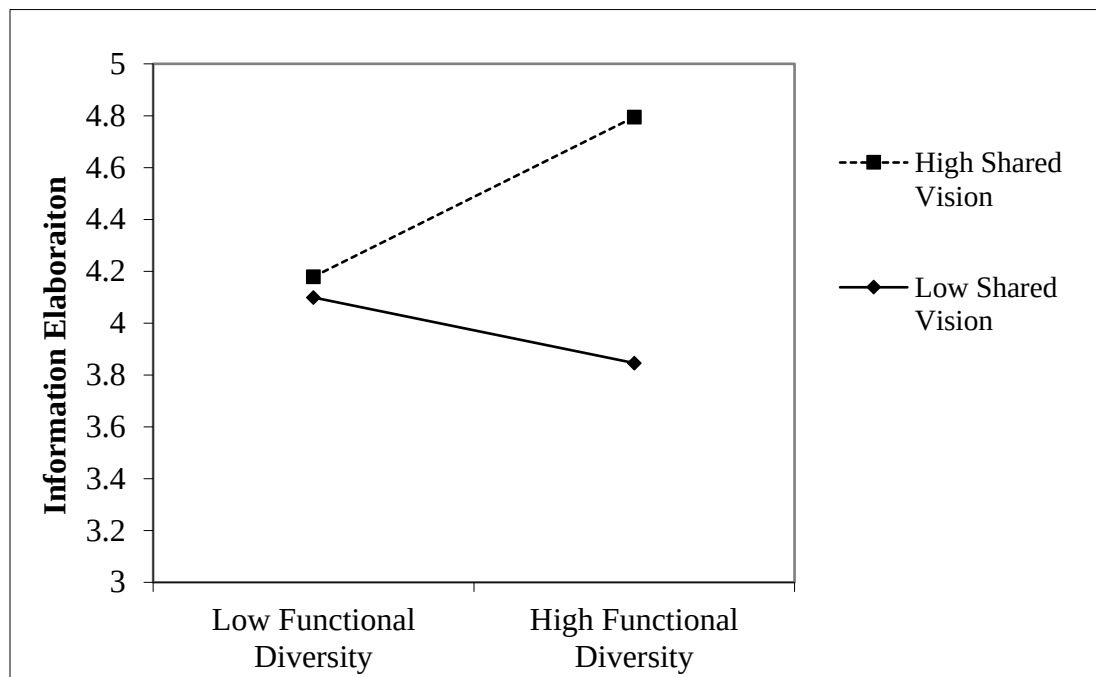


Figure 9 Moderating Effect of Shared Vision

As shown in the following Table 3, under the conditions of relatively high shared vision, the negative effect of functional diversity flipped into a positive effect. The Johnson-Neyman technique showed that the relationship between functional background diversity and information elaboration was significant for all cases with a mean-centered shared vision score of $-.14$ where 40% of the cases in the data set had values below, and

60% had values above this point, indicating that enough cases were within the significance region (Hayes, 2018a).

Table 3 Conditional Effects of Functional Diversity on Information Elaboration

	Effect	SE	<i>p</i>	95% CI
Low Shared Vision (-1 <i>SD</i>)	-.31	.11	<.01	-.53, -.10
Mean Shared Vision	.04	.08	.60	-.12, .21
High Shared Vision (+1 <i>SD</i>)	.40	.08	<.01	.24, .57
Boundary Spanning Treatment	-.12	.14	.39	-.39, .15
Boundary Spanning Control	.22	.08	<.01	.06, .39

Notes. *N* = 144. Values are standardized coefficients. CI = Confidence Interval.

The conditional effect of functional background diversity on information elaboration ranged from -.31 to .40 for a low and high shared vision, respectively, across the treatment and control group. While the boundary spanning treatment had a positive effect on information elaboration, its interaction with functional diversity surprisingly had a negative effect on information elaboration ($\beta = -.34$, $SE = .16$, $p < .05$). This indicates that the boundary-spanning role likely substitutes for the informational effect of functional diversity. In that vein, the boundary spanner as a source of information adds less (via scouting), resulting in a negative effect. Therefore, Hypothesis 2 and 3 are not supported. Overall, however, I found a positive effect of the boundary-spanning treatment on information elaboration, and teams in the treatment group performed at significantly higher levels of information elaboration.

As reported in Table 4, I found a similar data pattern of the conditional effects described for information elaboration for the conditional indirect effects of functional background diversity on team innovation. A bootstrap confidence interval for the indirect effects based on 20,000 bootstrap samples was entirely above zero for shared vision (.01,

.10) and entirely below zero for boundary spanning (-.12, -.01). Since the 95% confidence intervals did not cross over zero, these moderated mediation effects were statistically significant, supporting Hypothesis 6. The indices of moderated mediation are reported in Table 4.

Table 4 Conditional Indirect Effects of Functional Diversity on Team Innovation

	Effect	SE	95% CI
Low Shared Vision (-1 SD)	-.04	.02	-.10, -.01
Mean Shared Vision	.01	.01	-.01, .04
High Shared Vision (+1 SD)	.06	.03	.01, .12
Boundary Spanning Treatment	-.02	.02	-.05, .02
Boundary Spanning Control	.03	.02	.01, .08
Index of Moderated Mediation			
Shared Vision	.05	.02	.01, .10
Boundary Spanning	-.05	.03	-.12, -.01

Notes. N = 144. Values are standardized coefficients. CI = Confidence Interval.

In addition to the conditional indirect effects for values of each moderator, Table 5 reports the conditional indirect effects for all combinations of the moderators.

Table 5 Conditional Indirect Effects for all Combinations of the Moderators

Boundary Spanning	Shared Vision	Effect	SE	95% CI
Treatment	Low (-1 SD)	-.07	.03	-.14, -.01
	Mean	-.02	.02	-.06, .02
	High (+1 SD)	.03	.03	-.01, .10
Control	Low (-1 SD)	-.02	.02	-.07, .02
	Mean	.03	.02	.01, .08
	High (+1 SD)	.08	.04	.02, .16

Notes. N = 144. Values are standardized coefficients. CI = Confidence Interval.

Besides the conditional indirect effect of functional diversity on team innovation through information elaboration, I also found a significant direct effect of functional diversity ($\beta = .13$, $SE = .07$, $p < .05$), which was significantly moderated by boundary

spanning ($\beta = .27$, $SE = .13$, $p < .05$). As reported in Table 6, teams in the treatment group achieved higher levels of innovation than teams in the control group, providing further evidence that the boundary-spanning treatment worked.

Table 6 Conditional Direct Effects of Functional Diversity on Team Innovation

	Effect	SE	<i>p</i>	95% CI
Low Shared Vision (-1 <i>SD</i>)	.24	.09	<.01	.07, .41
Mean Shared Vision	.28	.07	<.01	.15, .41
High Shared Vision (+1 <i>SD</i>)	.31	.07	<.01	.02, .45
Boundary Spanning Treatment	.41	.11	<.01	.19, .62
Boundary Spanning Control	.13	.07	.05	-.00, .26

Notes. $N=144$. Values are standardized coefficients. CI = Confidence Interval.

To better discern the direct effect of the boundary spanning diversity interaction on team innovation, I examined a simple slope computation for the treatment ($\beta = .41$, $SE = .11$, $p < .01$) and control groups ($\beta = .13$, $SE = .07$, $p < .05$) as shown in Figure 10.

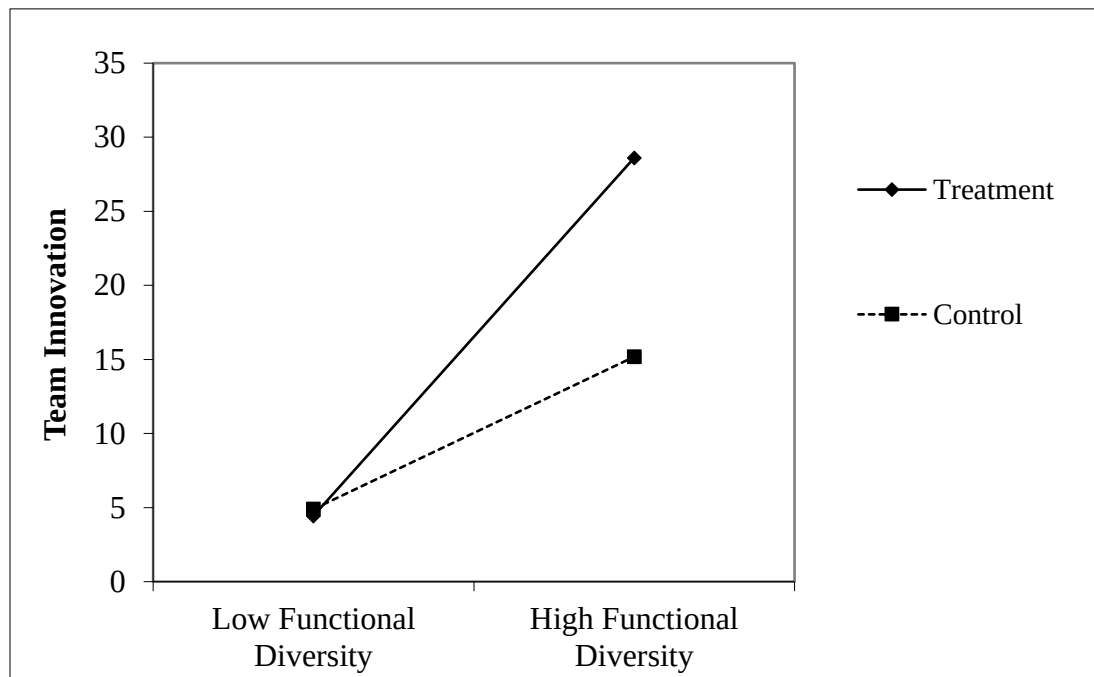


Figure 10 Moderating Effect of the Boundary Spanning Treatment

These results indicate that the treatment enhances the functional diversity effect on innovation in cross-functional teams. Therefore, Hypothesis 1 is supported. Similarly, teams with relatively high levels of shared vision achieved higher levels of team innovation. However, in contrast to boundary spanning, the interaction effect of shared vision and functional diversity had no significant effect on team innovation, and therefore Hypothesis 4 is not supported. The regression results without control variables are available in Appendix B, Table B1. While all control variables were statistically insignificant, the moderated mediation effects' overall data pattern without control variables was consistent, yielding a significant R^2 of 74%, which was four percentage points lower than in the analysis with control variables included.

Supplementary Analysis

Treatment-centric Moderated Mediation Analysis

In order to better understand the negative interaction effect of functional diversity and boundary spanning on information elaboration but the overall positive treatment effect, I conducted an ancillary post-hoc analysis that puts the boundary-spanning treatment at center stage. The treatment-centric research model shown in Figure 11 focuses on the boundary spanning treatment as an independent variable and includes serial mediation through shared vision and information elaboration with functional background diversity as a second-stage moderator. This way, I can explore and better understand the impact of boundary spanning on shared vision and the interplay with functional background diversity and information elaboration. By doing so, I analyze the direct and conditional effects of boundary spanning on team innovation through shared vision and information elaboration, moderated by functional diversity.

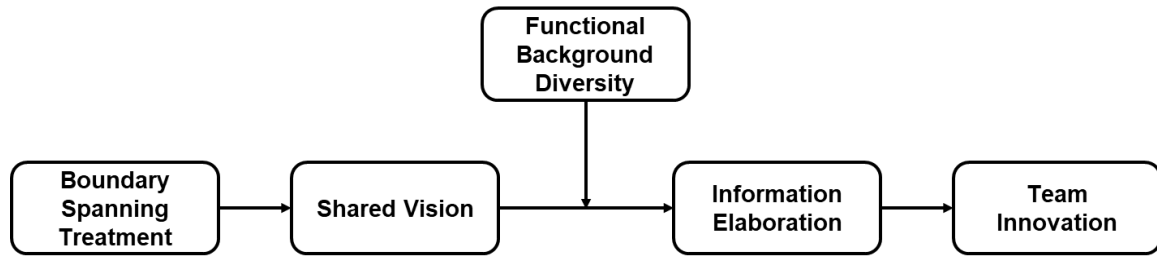


Figure 11 Treatment-centric Research Model

The following Table 7 reports the direct and conditional effects for the moderated serial mediation model. I utilized a custom version of PROCESS Model 91 (see Appendix C, Figure C1 with the treatment-centric path analysis results) to also include tests for moderation of the relationships between treatment and information elaboration, treatment and team innovation, as well as the relationship between shared vision and team innovation. I first regress shared vision on boundary spanning ($\beta = .39$, $SE = .16$, $p < .05$), and the control variables, which yielded a significant R^2 of 30%. Then information elaboration was regressed on boundary spanning ($\beta = .41$, $SE = .13$, $p < .01$), shared vision ($\beta = .48$, $SE = .07$, $p < .01$), functional diversity ($\beta = .22$, $SE = .08$, $p < .05$) and its interaction terms with boundary spanning ($\beta = -.34$, $SE = .16$, $p < .05$) and shared vision ($\beta = .36$, $SE = .05$, $p < .01$), and the control variables, which yielded a significant R^2 of 64%. Finally, team innovation was regressed on information elaboration ($\beta = .14$, $SE = .07$, $p < .05$), the boundary spanning treatment ($\beta = .19$, $SE = .11$, ns), shared vision ($\beta = .03$, $SE = .07$, ns), functional diversity ($\beta = .13$, $SE = .07$, $p < .05$) and its interaction terms with boundary spanning ($\beta = .27$, $SE = .13$, $p < .05$) and shared vision ($\beta = .03$, $SE = .04$, ns), as well as the innovation baseline ($\beta = .85$, $SE = .05$, $p < .01$) and the control variables, which yielded a significant R^2 of 78%.

Table 7 Direct and Conditional Effects of Boundary Spanning on Team Innovation

	Shared Vision	Information Elaboration	Team Innovation
Constant	.03 (.47)	.14 (.34)	-.51 (.27)
Boundary Spanning Treatment	.39 (.16)*	.41 (.13)**	.19 (.11)
Shared Vision	–	.48 (.07)**	.03 (.07)
Functional Diversity	–	.22 (.08)*	.13 (.07)*
Functional Diversity x Treatment	–	-.34 (.16)*	.27 (.13)*
Functional Diversity x Shared Vision	–	.36 (.05)**	.03 (.04)
Information Elaboration	–	–	.14 (.07)*
Team Innovation Baseline	-.10 (.08)	-.11 (.06)	.85 (.05)**
Team Size	-.08 (.09)	-.05 (.07)	-.09 (.05)
Executive Support	.34 (.08)**	-.09 (.06)	-.01 (.05)
Team Tenure	.25 (.08)**	.07 (.06)	-.02 (.05)
Work Experience	-.01 (.08)	-.05 (.06)	.04 (.05)
Consumer Industries	.02 (.48)	-.27 (.36)	.37 (.28)
Services Industries	-.49 (.52)	-.27 (.39)	.34 (.31)
Manufacturing Industries	-.30 (.48)	-.35 (.36)	.38 (.28)
Energy & Resources Industries	-.19 (.50)	-.36 (.37)	.44 (.29)
Financial Industries	-.72 (.54)	-.08 (.39)	.09 (.31)
Business Familiarity	-.16 (.08)*	.04 (.06)	-.06 (.05)
Annual Revenue	.02 (.09)	.02 (.06)	-.06 (.05)
COVID-19 Impact	-.50 (.35)	-.06 (.26)	-.18 (.21)
<i>R</i> ²	.30**	.64**	.78**
<i>F</i>	3.99**	12.36**	22.58**
Conditional Effect of Treatment on Information Elaboration			
	Effect (SE)	95% CI	<i>t</i>
Low Functional Diversity (-1 <i>SD</i>)	.75 (.23)**	.31, 1.20	3.33
Mean Functional Diversity	.41 (.13)**	.15, .68	3.08
High Functional Diversity (+1 <i>SD</i>)	.07 (.19)	-.31, .44	.36
Conditional Effect of Shared Vision on Information Elaboration			
	Effect (SE)	95% CI	<i>t</i>
Low Functional Diversity (-1 <i>SD</i>)	.12 (.07)	-.02, .26	1.72
Mean Functional Diversity	.48 (.07)**	.34, .62	6.80
High Functional Diversity (+1 <i>SD</i>)	.84 (.10)**	.65, 1.03	8.67

Notes. N = 144. Values are standardized coefficients (with standard errors).

* $p < .05$. ** $p < .01$. CI = Confidence Interval.

The regression results without control variables are available in Appendix B, Table B2. The Johnson-Neyman technique showed that the relationship between boundary spanning and information elaboration was significant for all cases with a mean-centered functional background diversity score of .41 where 56% of the cases in the data set had values below, and 44% had values above this point, indicating that enough cases were within the significance region. Similarly, the relationship between boundary spanning and team innovation was significant for all cases with a mean-centered functional background diversity score of .07, where 38% of the cases had values below, and 62% had values above this point. Table 8 reports the results of the treatment-centric moderated mediation model, including the conditional indirect effects.

Table 8 Treatment-centric Moderated Mediation Results

Conditional Direct Effect of Treatment on Team Innovation			
	Effect (SE)	95% CI	t
Low Functional Diversity (-1 SD)	-.08 (.19)	-.45, .29	-.43
Mean Functional Diversity	.20 (.11)	-.02, .41	1.79
High Functional Diversity (+1 SD)	.47 (.15)**	.18, .77	3.15
Conditional Indirect Effect of Treatment on Team Innovation through Shared Visio and Information Elaboration			
	Effect (SE)	95% CI	
Low Functional Diversity (-1 SD)	.01 (.01)	-.01, .03	
Mean Functional Diversity	.03 (.02)	.01, .07	
High Functional Diversity (+1 SD)	.05 (.03)	.01, .11	
Index of Moderated Mediation			
	Index (SE)	95% CI	
Functional Diversity	.02 (.01)	.01, .05	

Notes. N = 144. Values are standardized coefficients (with standard errors).

* $p < .05$. ** $p < .01$. CI = Confidence Interval.

As reported in Table 8, I found a positive moderated serial mediation effect of the boundary spanning treatment on team innovation through shared vision and information elaboration that is conditional on functional background diversity. A bootstrap

confidence interval for the indirect effect based on 20,000 bootstrap samples was entirely above zero for functional diversity (.01, .05). These results indicate that the treatment has a positive direct effect on team innovation that is conditional on functional diversity and an indirect effect through shared vision and information elaboration that is stronger for higher functional diversity levels. It is specifically notable that the boundary spanning treatment also positively affects shared vision, which is important for teams to benefit from functional background diversity. It is positively related to information elaboration only under a high shared vision. As illustrated in Figure 12, to better discern the nature of the significant interaction of functional diversity and shared vision, I examined a simple slope computation at “high” (1 *SD* above the mean; $\beta = .73$, $SE = .09$, $p < .01$) and “low” (1 *SD* below the mean; $\beta = .23$, $SE = .07$, $p < .01$) levels of functional diversity.

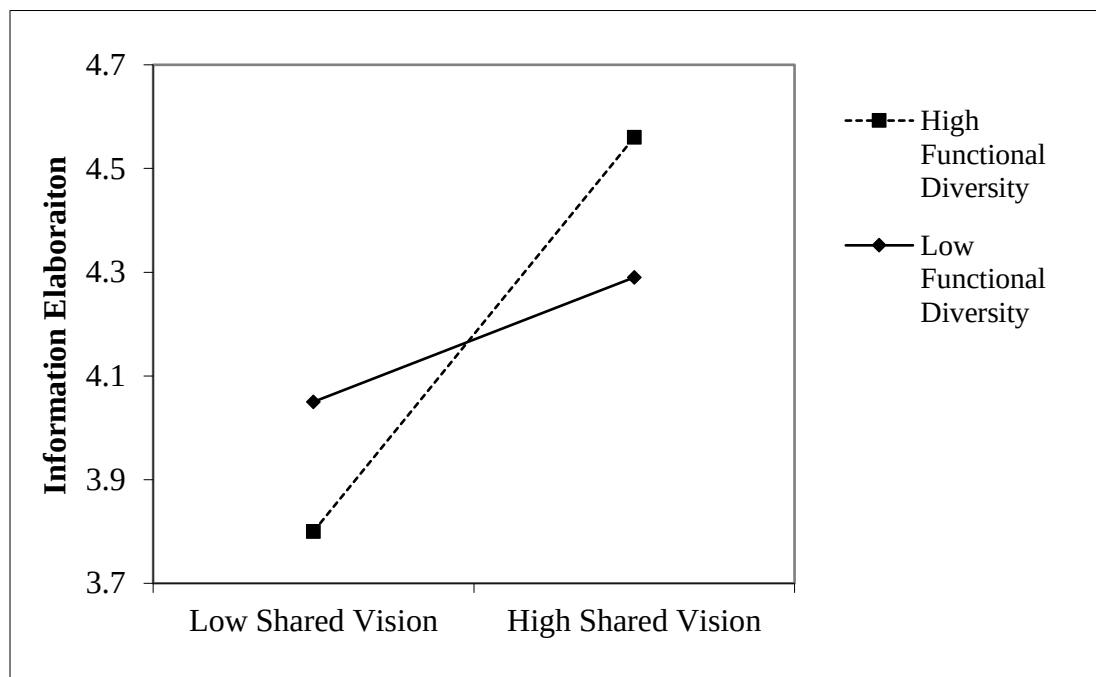


Figure 12 Interaction Effect of Functional Diversity and Shared Vision

These results emphasize the importance of shared vision showing that information elaboration benefits from functional diversity. However, teams with a low shared vision and high diversity perform at lower levels than teams with low diversity and low shared vision. As illustrated in the following Figure 13, to better discern the nature of the significant interaction of functional diversity and treatment on information elaboration, I examined a simple slope computation at “high” (1 *SD* above the mean; $\beta = .17$, $SE = .16$, *ns*) and “low” (1 *SD* below the mean; $\beta = .65$, $SE = .19$, $p < .01$) levels of diversity.

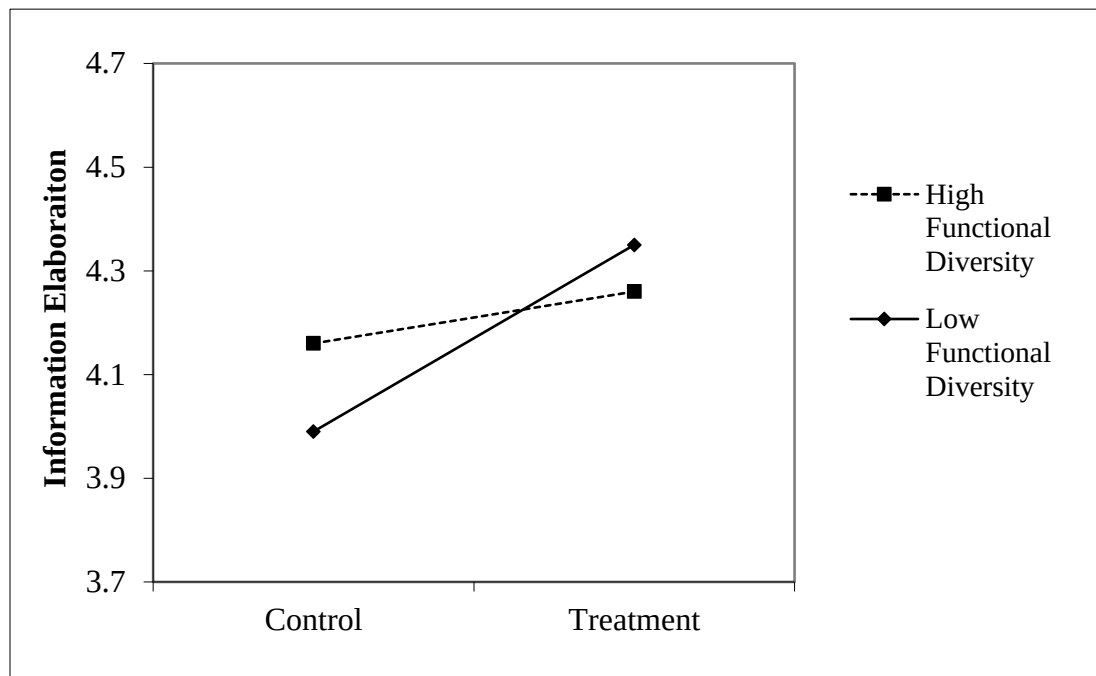


Figure 13 Interaction Effect of Functional Diversity and Boundary Spanning

These results show that teams in the treatment group perform at higher levels of information elaboration and emphasize the substitution effect of boundary spanning and functional diversity, which leads to relatively lower levels of information elaboration for teams with high functional diversity compared to low diversity in the treatment group.

Three-way Interaction Analysis

I also conducted an ancillary analysis to determine whether there is a three-way interaction effect of functional diversity, boundary spanning, and shared vision on team innovation. Table 9 reports the regression results for this three-way interaction analysis. I utilized PROCESS Model 3 to regress team innovation on functional diversity ($\beta = .13$, $SE = .06$, $p < .05$), boundary spanning ($\beta = .10$, $SE = .10$, ns), shared vision ($\beta = -.05$, $SE = .07$, ns), and the four interaction terms of functional diversity and the boundary spanning treatment ($\beta = .26$, $SE = .13$, $p = .05$), functional diversity and shared vision ($\beta = -.03$, $SE = .05$, ns), shared vision and treatment ($\beta = .10$, $SE = .11$, ns), as well as the product of functional diversity, shared vision, and treatment ($\beta = .36$, $SE = .10$, $p < .01$) while controlling for information elaboration ($\beta = .15$, $SE = .07$, $p < .05$) and the team innovation baseline ($\beta = .86$, $SE = .04$, $p < .01$), which yielded a significant R^2 of 77%.

Table 9 Three-Way Interaction Effect

Antecedent	Team Innovation		
	Coeff.	SE	p
Constant	-.16	.07	.02
Functional Diversity	.13	.06	.04
Boundary Spanning Treatment	.10	.10	.29
Functional Diversity x Treatment	.26	.13	.05
Shared Vision	-.05	.07	.44
Functional Diversity x Shared Vision	-.03	.05	.50
Shared Vision x Treatment	.10	.11	.38
Functional Diversity x Shared Vision x Treatment	.36	.10	<.01
Information Elaboration	.15	.07	.03
Team Innovation Baseline	.86	.04	<.01
$R^2 = .77$			
$F(9,134) = 49.85$, $p < .01$			

Notes. N = 144. Values are standardized coefficients.

The team innovation baseline and information elaboration explained 69% R^2 change ($F(2,141) = 155.09, p < .01$). 6% R^2 change was explained by functional diversity, boundary spanning, shared vision, and the two-way interactions ($F(6,135) = 5.41, p < .01$). The three-way interaction of functional diversity, shared vision, and boundary spanning explained 2% of the R^2 change ($F(1,134) = 12.80, p < .01$).

The positive significant coefficient of the three-way interaction indicated that the interaction of functional background diversity and shared vision on team innovation is more positive in the treatment group than in the control group. Since the interaction between functional diversity and shared vision differs significantly between treatment and control, it is reasonable that this interaction is not significant for team innovation by itself. Table 10 further reports the conditional effects of the interaction of functional background diversity and shared vision on team innovation for the boundary spanning treatment and control group. In contrast to the adverse effect in the control group (although not significant), the positive and significant treatment effect provides evidence that engagement teams with higher levels of functional diversity and shared vision that practice boundary spanning outperformed teams in the control group.

Table 10 Conditional Interaction Effect of Functional Diversity on Team Innovation

	Effect	<i>F</i>	<i>p</i>
Treatment	.33	12.32	<.01
Control	-.03	.45	.50

Notes. N = 144. Values are standardized coefficients.

To better discern the nature of the significant three-way interaction, I illustrated the impact on team innovation for high and low levels of functional background diversity and shared vision across the boundary-spanning treatment and control group in Figure 14.

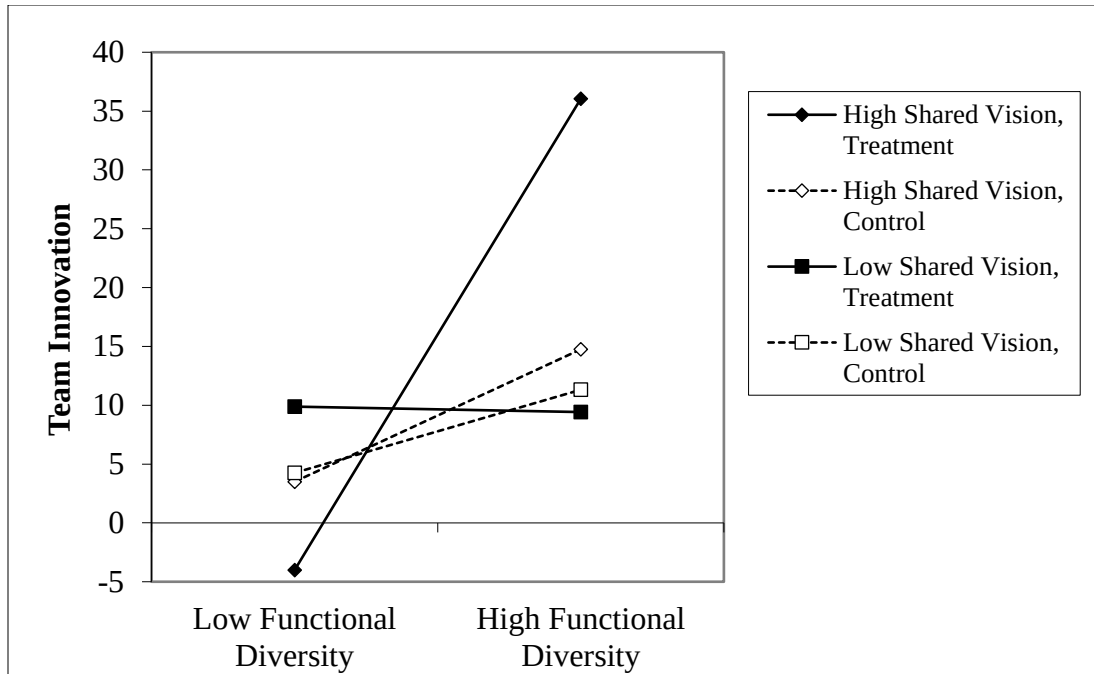


Figure 14 Three-way Interaction Effect

These results highlight the importance of the interplay of functional diversity, boundary spanning, and shared vision and demonstrate that teams in the treatment group are significantly more innovative only under the conditions of relatively high functional diversity and high shared vision. At lower levels of shared vision, functional diversity had no effect in the treatment group. The three-way interaction furthermore indicates that it is not only worth investing in functional diversity, boundary spanning, and shared vision in isolation since these factors enhance each other's effect and therefore should be developed in combination.

Findings and Interpretations

The boundary-spanning treatment worked and improved team innovation as measured via the use of software technology solutions. As shown in Figure 15, teams in the treatment group achieved on average 12 percentage points higher usage levels of their

software solutions during the measurement period. While the average usage rate in the treatment group increased from 44% to 56%, the control group usage rate decreased from 45% to 44% in the same time frame.

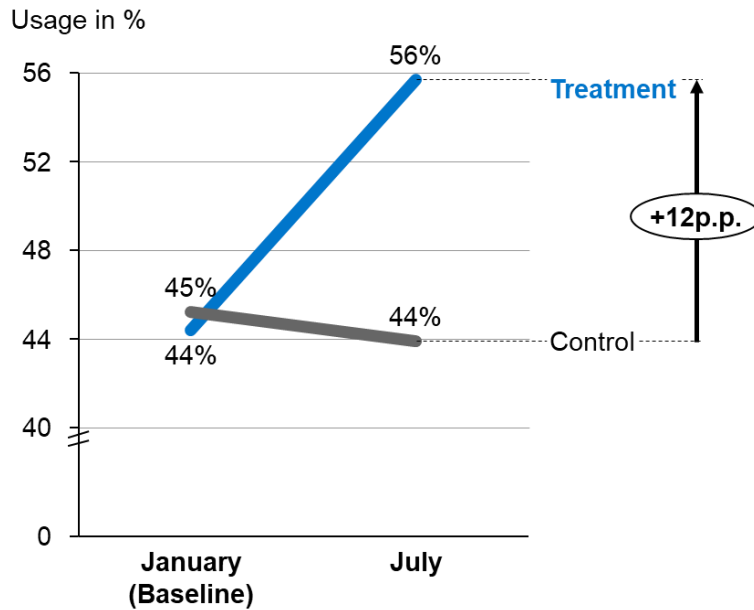


Figure 15 Result of the Team Innovation Experiment

The results of my analysis – presented in the following section – show an apparent positive effect of the boundary spanning treatment that increases information elaboration and team innovation. This raises the questions about which mechanisms, team processes, and conditions yielded this improvement.

One essential part of the answer is functional diversity. Functional diversity provides teams with the knowledge, information, expertise, and perspectives that help the team develop innovative ideas and overcome implementation challenges (van Knippenberg, 2017). As predicted in Hypothesis 1, I found a significant positive moderation effect of boundary spanning on the relationship between functional

background diversity and team innovation (see Table 2). As reported in Table 6, the effects of functional diversity on team innovation are significantly stronger in the treatment group. This data provides evidence for a positive relationship between functional diversity and team innovation that is stronger for teams that practice boundary spanning. With it, I confirmed the boundary-spanning treatment as facilitating condition and functional diversity as a critical informational resource required for team innovation.

The overall positive treatment effect set aside; the treatment has in a way a negative downside that reduces the positive influence of functional diversity on information elaboration. I found a significant negative moderation effect of the treatment on the relationship between functional diversity on information elaboration reported in Table 2, leading to the rejection of Hypothesis 2. While this is not necessarily negative in the sense that the treatment overall leads to more elaboration and innovation but because of the treatment, information elaboration is less driven by the team's functional background diversity, meaning the treatment reduces the informational effect of functional diversity. This unexpected finding contradicts the predicted positive pattern, where I suggested that functional diversity interacts with boundary spanning such that it is positively related to information elaboration, specifically because of additional information about the client and focus on the customer perspective. However, instead of the anticipated boost on information elaboration, I found that the treatment rather works as an alternative source of customer information (likely via scouting). Teams who received the treatment relied less on functional diversity as an informational resource.

Nevertheless, boundary spanning affects not only the quantity and how much the team elaborates (what I captured with my survey measure) but also the elaboration topics

and content quality. As confirmed in Hypothesis 1, creating this client focus leads to better use of functional diversity, and in that way, boundary spanning substitutes for information elaboration. As evidenced by the data, the boundary-spanning effect seems to be expressed more in the quality of information elaboration than in quantity.

For the same reasons and findings concerning Hypothesis 2, also Hypothesis 3 is rejected. Although I found a significant moderated mediation effect, the index of moderated mediation reported in Table 4 captures a negative effect indicating an overall less positive effect of functional diversity on team innovation through information elaboration conditional on boundary spanning. Despite this substitution effect, it is essential to note that teams in the treatment group operate at significantly higher information elaboration levels and delivered increased team innovation as measured by customer usage (see Figure 10). Furthermore, as reported in Table 5, the negative indirect effect flipped to a positive effect for relatively high levels of a shared vision.

The interaction of functional diversity and shared vision on team innovation was not statistically significant, rejecting Hypothesis 4. Despite partially significant effects for a range of shared vision (-1 to $+1$ *SD*) reported in Table 6 indicating that a shared vision and focus help to mobilize functional diversity resulting in increased team innovation for cross-functional teams, I cannot reject the null hypothesis in my sample population.

Contrary to Hypothesis 4, Hypothesis 5 is supported as I found evidence that functional diversity interacts with shared vision such that it is positively related to information elaboration. Meaning that the utilization of informational resources yielded from functional diversity in the team process of information elaboration – sharing, discussion, and integration of task-relevant information and perspectives – are motivated

by a shared vision and demotivated by the lack of shared vision. I found that in teams with a low shared vision, functional diversity has a negative effect. In teams with a high shared vision, functional diversity has a positive effect on information elaboration. This is of specific relevance because of the negative effect of functional diversity on information elaboration at relatively low shared vision levels. The negative effect at low levels of a shared vision is not entirely surprising since diversity in functional backgrounds also creates both task and personal conflict and may cause communication breakdowns. This, in turn, highlights the importance of the treatment to facilitate communication and coordination, shared vision as an idea of a valued outcome that provides the basis for action within the team, and potentially other team climate factors, such as psychological safety (Edmondson & Harvey, 2018), that motivate the use of informational resources.

As reported in Table 5, for a range of shared vision (-1 to $+1$ *SD*) across the treatment and control group, the relationship between functional diversity and information elaboration is conditional on boundary spanning and shared vision. While the tipping point from a negative to a positive effect differs between the treatment and control group, the data shows a positive relationship for teams with a relatively high shared vision.

As predicted in Hypothesis 6, I also found a moderated mediation effect where functional diversity indirectly influenced team innovation through its effect on information elaboration that was contingent on a shared vision. Shared vision significantly moderated the indirect effect of functional diversity on team innovation through information elaboration in such a way that these effects were more potent in teams with a shared commitment to common goals (see Table 4). As shown in Figure 9,

an increase in shared vision increases functional diversity's effect³ on information elaboration relative to teams with a low shared vision. As evidenced by the data, functional diversity affects team innovation directly and indirectly via information elaboration, which is the critical mediating team process of knowledge integration required to create synergies and foster team creativity and innovation (van Knippenberg, 2017). While Hypothesis 6 is supported as I found a significant moderated indirect path, I also have the unexpected finding that even with the mediation by information elaboration controlled for, there is still a direct effect not accounted for by information elaboration. This is where my supplementary analysis comes into play.

In that regard, it is essential to note that the boundary-spanning treatment, as found in the treatment-centric supplementary analysis shown in Table 7, not only positively influenced information elaboration and team innovation but also positively influenced shared vision, which increases the effect of functional background diversity and therefore is imperative for team innovation success. This could further explain why I found an indirect effect that captures how much the team elaborates and a direct effect where the treatment focused on what is important for the customer, directly feeding into customer usage. Finally, as reported in Table 9, the results of my supplementary three-way interaction analysis indicate that it is worth investing in functionally diverse teams with boundary-spanning roles and creating a shared team vision since all three factors

³ Consistent with the finding that team innovation benefits from functional background diversity, Balkundi and Harrison (2006) found that well-networked teams perform better and Baer (2010), building on Granovetter's strength-of-weak-ties theory, confirmed the importance of network diversity as critical factor for creative outcomes. In my analysis, similarly to functional diversity, I found a significant moderated mediation effect of team network diversity on team innovation through information elaboration that is conditional on shared vision and boundary spanning. Network diversity however, was not significant when analyzed together with functional diversity, indicating that they are substitutes for one another. Functional and network diversity were significantly correlated ($r = .38, p < .01$).

enhance each other's effect on team innovation. Interestingly, the three-way interaction analysis (see Figure 14) shows that teams with an above-average shared vision and low functional diversity are generally less likely to drive innovation than teams with a low shared vision and low functional diversity. Again, this finding speaks to the importance of information elaboration and boundary spanning since shared vision and functional diversity alone do not deliver the desired innovation success. The boundary-spanning treatment creates a focus on serving the customer and amplifies the effects of functional diversity, shared vision, and information elaboration on team innovation. As evidenced by the data analysis in this study, a team's capability to generate technology-based ideas new for the adopting customer and putting these ideas into practices such that they achieve desired results benefits from functional diversity and is advanced through boundary spanning and shared vision.

In summary, teams with a high diversity of functional backgrounds that operate with a shared commitment to common goals and engage in continuous information elaboration by orchestrating proactive engagements innovate better with their customers. My conditional process model explained 78% of the variance in team innovation change starting from the January baseline till the end of the experiment. 17% R^2 change was explained by the main effect of functional background diversity on team innovation through information elaboration that was conditional on boundary spanning and shared vision. Although all control variables were non-significant, the inclusion added an additional four percentage points to the overall team innovation explanation.

CHAPTER 5: DISCUSSION

In this chapter, I first recap the aim of my study briefly. Then, I dig into my findings' theoretical contributions and address practical and managerial implications focusing on recommendations and how to implement the results. Afterward, I outline the limitations and shortcomings of my research design and how future research could address these. Finally, I conclude with a punchline summary of my study, the key findings, and their importance.

The goal of my study was to better understand the role of boundary spanning for team innovation success and develop a deeper level of understanding under what conditions customer engagement teams in technology companies are likely to benefit from functional background diversity in pursuing digital innovation challenges together with their clients. Specifically, exploring the effects of shared vision and a boundary-spanning treatment with a field experimental research design. The central thesis is that team innovation benefits from informational resources through its relationship with information elaboration and engenders team innovation success only if teams operate in a customer-centric climate for innovation with a common goal shared by the team. In this context, the boundary-spanning treatment was designed to bring in information about the client and create a focus on the client's perspective, resulting in increased software solution use.

Theoretical Contributions

I applied van Knippenberg's (2017) knowledge integration / team climate model of team innovation in my research. My study results provide supporting evidence for the proposed moderated mediation effect of functional background diversity – as an

informational resource – on team innovation through information elaboration that is conditional on shared vision as one key factor of the broader team climate construct. Most notably, by integrating the team climate and knowledge integration perspectives and combining it with the notion of boundary spanning, I offer a rich and novel investigation of the role of boundary spanning for idea implementation in customer engagement teams. My study contributes to the diversity and information elaboration theories and research by revealing that in the information elaboration process, we need to account for the moderating role of boundary spanning and shared vision for functional diversity effects on team innovation. My research findings are consistent with significant reviews (e.g., van Knippenberg, 2017) and meta-analyses (van Dijk et al., 2012) in suggesting that functional diversity can have positive and negative effects on team innovation. Specific moderating influences are required to unlock the potential of functional diversity. Both boundary spanning and shared vision are such moderators. My study provides evidence that functional diversity positively affects team innovation through information elaboration in teams with a shared vision of their work.

When team members lack a shared vision for their work, then the categorization process prevails over information elaboration. In such a situation, functional diversity negatively affects teamwork, defined as sharing, discussing, and integrating task-relevant information (van Knippenberg et al., 2004). By contrast, when team members share a common, higher-order goal within the team, information elaboration becomes dominant so that cross-functional teams create synergies from the different perspectives and integrate their knowledge, which then engenders innovative outcomes. This finding is consistent with Somech and Drach-Zahavy (2013) and Pearce and Ensley (2004), who

arrived at similar conclusions for primary care teams and product and process innovation teams, respectively. These findings provide answers to my first research question explaining how and under what conditions teams are more likely to benefit from diverse informational resources in pursuing innovation challenges.

Focusing on the company experiment, I explored the impact of functional background diversity on team innovation as impacted by the boundary-spanning treatment, providing evidence that functional diversity is positively related to team innovation. This relationship is stronger for teams that practice boundary spanning. I also provide evidence of an interaction in which boundary spanning moderates the functional diversity-information elaboration relationship, impacting team innovation. However, in contrast to my prediction, I found a negative moderation effect. While several studies (e.g., Somech & Khalaili, 2014) found negative effects of specific forms of boundary spanning, in the vein of buffering or boundary-tightening activities, this must not be confused with the evidence I found. In teams that practice boundary spanning, I found evidence that information elaboration is less driven by the team's functional background diversity. The treatment reduced the informational effect of functional diversity and boundary spanning substituted for information elaboration. Although Hypothesis 2 and 3 were not supported because of the negative moderation effect, it seems that the boundary spanning effect is presumably more expressed in the quality of information elaboration than in quantity. This explanation is consistent with the theory and other empirical evidence. For example, Somech and Khalaili (2014) found a positive relationship between team functional heterogeneity and team innovation mediated through boundary-spanning activities, indicating that boundary spanning, specifically the task-coordination

activities, can substitute for information elaboration as a key process for knowledge integration. My supplementary analysis further provides evidence that boundary spanning brings in information about the customer, focuses on the customer, and influences what the team elaborates. However, as shown with the support of Hypothesis 1, teams that practice boundary spanning can also develop and implement innovative solutions with their clients without extensive information elaboration in the form of detailed discussions of perspectives. This is consistent with Sung and Choi's (2012) research and Nadkarni and Narayanan (2007), who specifically point out that teams build on past experiences and procedures that are known to be working. Since most of the teams in my sample worked on incremental innovations with available best practices instead of radical or disruptive innovations, this finding is consistent with the broader body of knowledge (Alexander & van Knippenberg, 2014). Interestingly and in contrast to the boundary spanning interaction effect, the shared vision did not significantly interact with functional diversity to create a stronger relationship with team innovation (Hypothesis 4). While a shared vision and focus should help to mobilize functional diversity resulting in increased team innovation (Fay et al., 2006), it emphasizes the role of information elaboration as a key team process that enables teams to leverage their informational resources to drive innovation. Specifically in more complex service engagements with technology providers that require business and IT alignment (Hoever et al., 2012; Sabherwal et al., 2001). However, it is also possible that I have a restriction of range in functional diversity and shared vision in my sample teams. Meaning that all teams have enough of an understanding of what they are supposed to do together and that the variation of a shared vision between teams is within a restricted range of relatively high levels. Given that my

sample teams are customer-facing and given the quality of people on the teams, I do not have teams with no shared vision. Therefore, I compare relatively speaking rather moderate to high levels of shared vision, and because of this, it may suppress effects that exist. Similarly, my study departs from many empirical studies of diversity that include completely homogeneous teams and are often restricted because the included teams are not very diverse. On the contrary, the teams in my data set are at the higher end of functional diversity, with almost none being homogeneous in functional backgrounds within the team. Therefore, it is possible that some of these limitations in range suppress the theoretically expected effects. This phenomenon also needs to be considered when interpreting the three-way interaction effect. Although I found a positive significant three-way interaction effect, this analysis relies on the less frequent occurrence of low shared vision in the treatment group even when treatment tends to increase shared vision, pointing to potential unmeasured covarying variables. On the contrary, the treatment-centric analysis of the serial mediation through a shared vision and information elaboration with functional diversity as second-stage moderator is more robust. It provides novel insight that boundary spanning (via ambassadorial activities) increases shared vision. Whereas my focal analysis suggested that boundary spanning teams make less use of informational resources from the functional diversity in information elaboration, my supplementary analysis and qualitative insight suggest that boundary spanning creates a focus on the client through a shared vision and increases information elaboration. This insight also links to the related research on cooperative goals and shared identity that highlights the importance of teamwork, focusing on shared goals as a requirement for teams to be innovative (van Knippenberg, 2017). Hülshager and

colleagues (2009), to that point, provide meta-analytic evidence for the positive influence of team cohesion and goal interdependence on team innovation. Tjosvold, Tang, and West (2004) show that team innovation benefits from cooperative goals mediated through team reflexivity.

Beyond previous research focused on a shared vision and goals (e.g., Lisak et al., 2016), the notion of team reflexivity also, in a way similar to boundary spanning, connects to the notion of design thinking, which helps to focus on the user's needs and brings in the user perspective to iteratively ideate and implement solutions that address critical business issues (Cousins, 2018; Venkatesh et al., 2012). I have theorized and empirically validated a model of how the relationship between functional diversity and team innovation is moderated by boundary spanning, addressing my second research question. Here my contribution is the finding that illustrates how boundary spanning enhances team innovation for cross-functional customer engagement teams.

Managerial Implications

Most notably, support for my research model suggests that technology providers in particular, or organizations in general, benefit from boundary-spanning team activities and cross-functional team setups to drive customer innovation. As evidenced by the field experiment results, the boundary-spanning treatment had a significant positive effect on team innovation as measured by customer solution usage and on information elaboration within the team and the team's shared vision. These are critical for teams to leverage diverse insights and perspectives in achieving business outcomes for their customers. Driving customer success when many organizations struggle with their digital transformation initiatives is critical for both the client's sustained competitive advantage

in the industry and the technology provider's ability to retain customers and maintain profitable recurring revenue streams. In the company experiment, it was found that customers in the treatment group achieved on average 12% higher usage of their software solutions during the measurement period, and customers with increasing usage rates show 14% in revenue growth. In contrast, customers with stable or declining utilization show no revenue growth. Based on this experimental evidence, the company increases its investments in customer success and boundary-spanning team roles to drive customer lifetime value.

Furthermore, the vast academic debate of research designs with mediation models shows that such models are efficient because of the intended causal relationships providing step-by-step guidance for crucial factors and focus areas in the process to achieve the desired outcomes (e.g., Harvey et al., 2019). In that sense, functional background diversity should be considered a critical factor for team composition when creating account teams. It is one of the primary sources of knowledge, information, expertise, and perspectives that help teams develop innovative ideas and overcome implementation challenges. Leaders and executives should consider investments in boundary-spanning to facilitate knowledge integration and leverage the team's diverse perspectives and expertise to benefit the customer. Knowledge integration in the form of information elaboration in cross-functional teams is an essential step in the team innovation process recommended as a focus area for leaders. Depending on the context, boundary spanners can take on an orchestration role for the transformation effort and increase the overall chances for success. In that vein, companies should also train their account teams for joint success planning and create a shared vision since the idea of a

valued outcome represents a substantial motivating force at work. It is critical that the work is personally important to team members and that all members believe in their work's purpose. Although it is undoubtedly intuitive and straightforward that teams should have motivating goals, my supplementary analysis shows that boundary spanning, specifically through ambassadorial activities, positively affects the team's shared vision. It also creates a client focus in information elaboration. By considering all these benefits together, executives and leaders have a strong case to invest in boundary-spanning for teams pursuing innovation challenges.

Finally, I found a significant three-way interaction effect of boundary spanning, functional diversity, and shared vision. Although there is a risk of unmeasured variables that covary in the three-way analysis (specifically because low shared vision is rare in the treatment group), it is valuable to note for executives that not only is it worth investing in these factors in isolation, but they should be developed in combination to create highly innovative work teams. In summary, on their relentless mission to develop the perfect team, executives and team leaders should put boundary spanning on the agenda in order to achieve new levels of customer-centricity and facilitate innovation management in cross-functional groups.

Limitations and Future Research

In contrast to prior team innovation studies, which mostly analyzed teams with perceptual outcome measures, I analyzed cross-functional customer engagement teams with an objective measure of team innovation by evaluating customer usage as the ultimate measure of innovation success. While this addressed some limitations of previous research, my research suffers from its own shortcomings, which I elaborate on

in the following sections. First, the company field experiment was not without challenges. For example, a few customers rejected the treatment, and the COVID-19 crisis emerged in the middle of the experiment. However, despite the boundary-spanning treatment working out very well, functional background diversity relied on natural variation given the quasi-experimental design. Therefore, I do not know to what extent other team characteristics that covary with functional diversity cause its effect. Ideally, future research would manipulate functional background diversity to reduce the potential for alternative interpretations.

Second, my survey measure of information elaboration captured how much teams elaborate but does not provide insights into their elaboration content. While in my context boundary spanning is presumably more about what is important for the client, for some interpretations, I am relying on the unmeasured variable of the content of elaboration. Therefore, future research would ideally provide more information about team information elaboration quality and content to substantiate my interpretations further. Such an analysis would also benefit from a broader perspective on different types of informational resources. For example, studying social network diversity (e.g., Hirst, van Knippenberg, Zhou, Quintane, & Zhu, 2015; Reagans & Zuckerman, 2001) and potential interdependencies of functional diversity and network diversity where, at least in the context of customer engagement teams, one may argue that increased functional diversity leads to increased network diversity. Like functional background diversity, team networks are considered critical informational resources that help teams drive innovative outcomes if leveraged in the right way (Balkundi & Harrison, 2006; van Knippenberg, 2017). The notion of increased diversity leading to an increased performance also raises

the question for a more nuanced analysis of the effects of diversity by, for example, looking at the effects of adding specific functional backgrounds to the team (van Knippenberg, Dawson, West, & Homan, 2010) and better understanding potential curvilinear relationships where diversity is potentially only better up to a specific point (e.g., Tzabbar & Vestal, 2015).

Third, I only partially cover the two main themes of knowledge integration and team climate in the broad field of team innovation research (van Knippenberg, 2017). Analyzing the effects of further factors of the team climate inventory (Anderson & West, 1998), such as psychological safety (Edmondson, 2003; Harvey et al., 2019), would provide a more comprehensive view on how the use of informational resources in the team process of information elaboration can be stimulated. In that regard, the team creativity and innovation body of knowledge would also benefit from further consolidation of potentially overlapping constructs. This includes elements of the team climate inventory, such as participative safety, which is related to psychological safety⁴ and trust, or shared vision that is related to cooperative goals and shared identity (e.g., Tjosvold et al., 2004). For example, van Knippenberg and Hoever (in press) also address the latest developments and importance of team leadership in their literature review. Especially for customer engagement teams working with clients on strategic digital transformation initiatives, it would be fascinating to understand better how leadership can impact team innovation by motivating knowledge integration. D’Innocenzo, Mathieu, and

⁴ I also measured psychological safety which is defined as shared belief that the team is safe for interpersonal risk taking (Harvey et al., 2019) but I had the measurement problem that it overlapped with information elaboration. Since I was more interested in the team process of information elaboration which is more to the point of what is actually done in the team (vs. the individual feeling or perception of how freely one can contribute), I dropped psychological safety from my analysis.

Kukenberger (2016) provide meta-analytic evidence for a positive relationship between shared leadership and team performance. Kukenberger and D’Innocenzo’s (2020) research shows that shared leadership benefits from functional diversity when teams operate in a cooperative climate and consider the team dynamics over time. Li, She, and Yang (2018) furthermore found that innovation benefits from the diversity of expertise through perspective-taking and that this relationship is moderated by paradoxical leadership. Additionally, Zacher and Rosing (2015) show in their research that ambidextrous leadership with stimulation for divergence and convergence has a positive effect on team innovation.

Lastly, given the specific nature of customer engagements and my U.S.-based sample, my findings may be generalizable only to such teams involved in projects driving customer success with enterprise software solutions considering the specific cultural norms across North America.

Conclusion

The results of my research, withstanding some limitations, provide a detailed understanding of the role of boundary spanning for team innovation and emphasize the importance of shared vision and the information elaboration process in cross-functional teams. My study advances information elaboration theory and shows that the relationship between functional diversity and team innovation is mediated by information elaboration and is contingent on a shared vision and boundary spanning. I found that team innovation benefits from functional diversity and boundary spanning only in teams with a relatively high shared vision. While functional diversity and boundary spanning can increase information elaboration, boundary spanning can also substitute for information

elaboration and not only affects how much teams elaborate but also about what they elaborate, meaning what is vital for the client. By bringing in information about the client and focusing on the customer perspective, boundary spanning leads to better use of functional diversity that directly feeds into client usage. My research provides experimental evidence that boundary-spanning team activities foster knowledge integration and enhance team innovation. Technology providers seeking to enhance customer team innovation should create teams with diverse functional backgrounds and focus on boundary-spanning activities since that not only puts the customer in the center of all team activities but also enhances the shared higher-order goals of the team to deliver for their clients. For customer engagement teams to benefit from informational resources in general and functional background diversity of team members in particular, they need to integrate diverse information and invest in boundary spanning and a shared commitment to common goals since these factors enhance each other's effect on team innovation. Therefore, my findings contribute actionable insight into how to enable cross-functional engagement teams to drive customer success appropriately.

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APPENDIX A: Measures

Boundary Spanning

I reused the following items from Faraj & Yan (2009, p. 611) to measure boundary spanning:

- Encourage its members to try to influence important stakeholders elsewhere in and/or beyond the division on behalf of the team and its work?
- Encourage its members to solicit information and resources from elsewhere in and/or beyond the division?
- Value team members for making use of their relationships with others on behalf of the team?
- Depend upon information and resources actively solicited by team members?

Survey respondents were asked to indicate their agreement to these items on a 5-point scale ranging from “strongly disagree” to “strongly agree.”

Shared Vision

For shared vision, I reused the following measurement items from Anderson & West (1998, p. 241) as defined in their team climate inventory:

- How clear are you about what your team’s objectives are?
- To what extent do you think they are useful and appropriate objectives?
- How far are you in agreement with these objectives?
- To what extent do you think your team's objectives are clearly understood by other members of the team?
- To what extent do you think your team's objectives can actually be achieved?
- How worthwhile do you think these objectives are to the organization?

Team members were asked to rate each item for their team on a 7-point scale ranging from “not at all” to “completely.”

Psychological Safety

I measured psychological safety with four items defined by Harvey and colleagues

(2019):

- In this team, it is easy to speak up about what is on your mind
- People in this team are usually comfortable talking about problems and disagreements
- People in this team are eager to share information about what does and doesn't work
- If you make a mistake in this team, it is often held against you (reverse scored)

Each item was rated for the team on a 7-point scale ranging from “strongly disagree” to “strongly agree.”

Information Elaboration

I measured information elaboration with six items defined by van Knippenberg and

colleagues (2019): “In our team...”

- Members complement each other by sharing their knowledge
- We discuss alternatives before making decisions
- We combine information from different members to gain a better understanding of the task
- We combine the input of different members
- We consider the implications of the combination of different members' viewpoints
- We exchange different ideas and perspectives on the task

As part of the survey, respondents were asked to indicate their agreement to these items within their team on a 5-point scale ranging from “strongly disagree” to “strongly agree.”

APPENDIX B: Regression Results without Controls

Table B1 Direct and Indirect Effects of Functional Diversity on Team Innovation through Information Elaboration, moderated by Boundary Spanning and Shared Vision

	Info. Elaboration			Team Innovation			
	Antecedent	Coeff.	SE	p	Coeff.	SE	p
Constant		-.09	.08	.26	-.14	.07	.05
Functional Diversity		.25	.08	<.01	.08	.06	.20
Boundary Spanning Treatment		.32	.12	.01	.15	.13	.15
Functional Diversity x Treatment		-.38	.15	.01	.32	.04	.01
Shared Vision		.47	.06	<.01	.06	.06	.35
Functional Diversity x Shared Vision		.36	.04	<.01	.05	.04	.21
Team Innovation Baseline		-.08	.05	.15	.85	.04	<.01
Information Elaboration		—	—	—	.12	.07	.09
		$R^2 = .62$			$R^2 = .74$		
		$F(6,137) = 37.44, p<.01$			$F(7,136) = 56.24, p<.01$		

Notes. N = 144. Values are standardized coefficients. Control variables are not included.

Table B2 Direct and Conditional Effects of Boundary Spanning on Team Innovation through Shared Vision and Information Elaboration, moderated by Functional Diversity

	Shared Vision	Information Elaboration	Team Innovation
Constant	-.19 (.12)	-.09 (.08)	-.14 (.07)*
Boundary Spanning Treatment	.36 (.16)*	.32 (.12)*	.15 (.10)
Shared Vision	–	.47 (.06)**	.06 (.06)
Functional Diversity	–	.25 (.08)**	.08 (.06)
Functional Diversity x Treatment	–	-.38 (.15)*	.32 (.13)*
Functional Diversity x Shared Vision	–	.36 (.04)**	.05 (.04)
Information Elaboration	–	–	.12 (.07)
Team Innovation Baseline	-.15 (.08)	-.08 (.05)	.85 (.04)**
R^2	.05*	.62**	.74**
F	4.03*	37.44**	56.24**

Conditional Effect of Treatment on Information Elaboration at Values of Functional Diversity			
	Effect (SE)	95% CI	t
Low Functional Diversity (-1 SD)	.59 (.18)**	.24, .94	3.34
Mean Functional Diversity	.22 (.12)	-.02, .45	1.83
High Functional Diversity (+1 SD)	.05 (.14)	-.23, .34	.37

Conditional Effect of Shared Vision on Information Elaboration at Values of Func. Diversity			
	Effect (SE)	95% CI	t
Low Functional Diversity (-1 SD)	.22 (.06)**	.11, .34	3.80
Mean Functional Diversity	.57 (.06)**	.45, .70	8.88
High Functional Diversity (+1 SD)	.73 (.08)**	.58, .87	9.68

Notes. N = 144. Values are standardized coefficients (with standard errors).

* $p < .05$. ** $p < .01$. CI = Confidence Interval. Control variables are not included.

APPENDIX C: PROCESS Custom Model 91

SPSS PROCESS command for supplementary treatment-centric moderated serial mediation analysis with standardized variables and custom wmatrix:

PROCESS y=ZTIC7/x=Treatment/m=ZTSDVN ZTINEN/w=ZCTFBDYBN/model=91
/cov=ZTICB zctsize ztexecsp zttenure ztwkexp ic_consumer ic_service ic_discrete ic_enr
ic_finance zcustomer_since_days zrev_2019
covid19/wmatrix=0,1,1,1,0/boot=20000/decimals=F10.2/jn=1/moments=1.

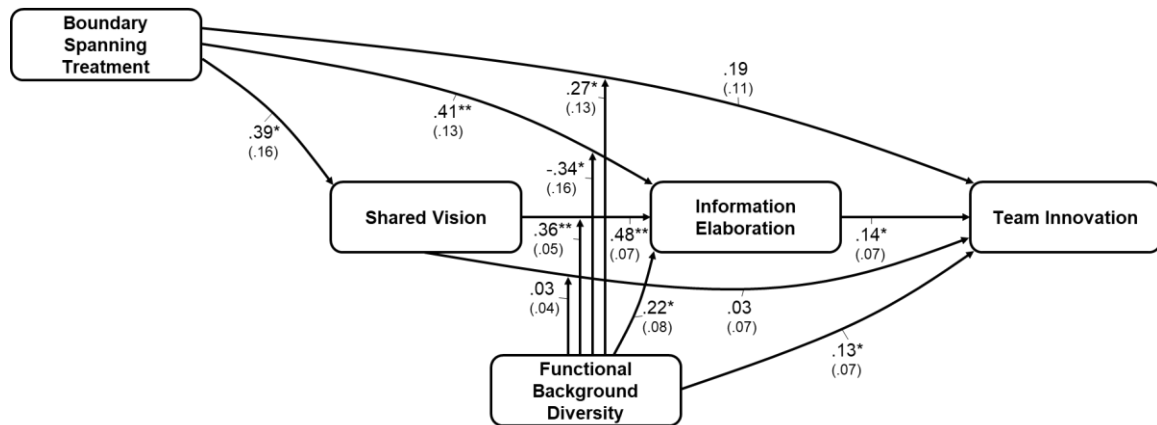


Figure C1 Treatment-centric Path Analysis Results

Notes. N = 144. Values are standardized coefficients (with standard errors).

* $p < .05$. ** $p < .01$.

